Improved Statistical Inference for Expectile-Based Risk Measures

An Optimal Pooling Approach in the Heavy-Tailed Regime

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Abstract

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Background

2.1 Regular variation and the second-order condition

Throughout the thesis we work with a real-valued random loss X that has continuous distribution function F and survival function $\overline{F} = 1 - F$. Given an iid sample $X_1, \ldots, X_n$ from F, the corresponding order statistics are denoted by $X_{1:n} \leq X_{2:n} \leq \cdots \leq X_{n:n}$.

Heavy-tailed asymptotics are governed by the language of regular variation; the reference is Haan and Ferreira (2006, Appendix B.1).

Definition 2.1 (Regular variation)**.** A Lebesgue measurable function $f : (0, \infty) \to \mathbb{R}$ that is eventually positive is *regularly varying at infinity with index* $\alpha \in \mathbb{R}$, written $f \in \mathrm{RV}_\alpha$, if for every $y > 0$,

$$\lim_{t\to\infty} \frac{f(ty)}{f(t)} \;=\; y^\alpha.$$

The functions in RV_0 are called *slowly varying*.

The first-order condition under which all of our asymptotics are taken asserts that F has a Pareto-type tail.

Condition 2.2 (First-order condition)**.** There exists $\gamma > 0$ such that $\overline{F} \in \mathrm{RV}_{-1/\gamma}$, or equivalently the *tail quantile function*

$$U(t) \;\coloneqq\; \inf\Big\{x \in \mathbb{R} \,:\, \tfrac{1}{\overline{F}(x)} \geq t\Big\}, \qquad t > 1,$$

satisfies $U \in \mathrm{RV}_\gamma$.

Throughout, for a continuous distribution function F, we write

$$Q_X(\tau) \;\coloneqq\; F^{\leftarrow}(\tau) = \inf\{x \in \mathbb{R} : F(x) \geq \tau\}, \qquad 0 < \tau < 1.$$

With the tail quantile $U(t) = F^{\leftarrow}(1 - 1/t)$, $t > 1$, this gives

$$Q_X(\tau) = U((1-\tau)^{-1}), \qquad Q_X(1-p) = U(1/p).$$

Condition 2.2 is the Fréchet max-domain condition with extreme-value index γ (Haan and Ferreira 2006, Theorem 1.2.1); standard examples include Pareto, Burr, Student-t, Fréchet and F-distribution tails (Haan and Ferreira 2006, §2.6). For normal limits we use the standard second-order refinement.

Condition 2.3 (Second-order condition $\mathcal{C}_2(\gamma, \rho, A)$)**.** There exist $\gamma > 0$, a second-order parameter $\rho \le 0$, and a measurable auxiliary function A of constant sign with $A(t) \to 0$ as $t \to \infty$, such that for every $y > 0$,

$$\lim_{t\to\infty} \frac{1}{A(t)} \left[\frac{U(ty)}{U(t)} - y^{\gamma} \right] \;=\; y^{\gamma}\, \frac{y^{\rho}-1}{\rho},$$

where the right-hand side is read as $y^{\gamma} \log y$ when $\rho = 0$.

The function $|A|$ is itself regularly varying with index ρ (Haan and Ferreira 2006, Theorem 2.3.9). Since A has constant sign eventually, signed ratios $A(ty)/A(t)$ converge to y^{ρ}; the rate A is the source of the second-order bias terms below.

Condition 2.4 (Finite first-moment range)**.** In addition to Condition 2.2, the extreme-value index satisfies $\gamma \in (0, 1)$ and the negative part $X^- = \max(-X, 0)$ satisfies

$$\mathbb{E}\, X^- < \infty.$$

Under Condition 2.2, $\gamma < 1$ gives $\mathbb{E}\, X^+ < \infty$; together with the displayed lower-tail condition this is the finite-mean regime for the quantile-based expectile approximation.

2.2 Tail-index and extreme-quantile estimation

We collect the two classical estimators used throughout the thesis: the Hill estimator of γ and the Weissman extrapolation for extreme quantiles. The textbook reference is Haan and Ferreira (2006, Ch. 3–4).

2.2.1 The Hill estimator

Given an iid sample $X_1, \ldots, X_n$ from F and an integer $k = k_n \in \{1, \ldots, n-1\}$, the *Hill estimator* (Hill 1975) of γ based on the $k+1$ top order statistics is

$$\hat{\gamma}_n(k) \;=\; \frac{1}{k} \sum_{i=1}^{k} \log\!\left(\frac{X_{n-i+1\,:\,n}}{X_{n-k\,:\,n}} \right). \tag{2.1}$$

Since $U(t) \to \infty$, $X_{n-k:n} > 0$ with probability tending to one for intermediate k; log-based estimators are understood on this event. For tail-index estimation one may also shift the data, but shifts must be undone for quantile or expectile estimation. The consistency result below is Haan and Ferreira (2006, Theorem 3.2.2); the normal limit is Haan and Ferreira (2006, Theorem 3.2.5) when $\rho < 0$, with the $\rho = 0$ endpoint covered by the $m = 1$ specialisation of Daouia, Padoan, et al. (2024, Theorem 1).

Theorem 2.5. *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$. Let $k = k_n \to \infty$ and $k/n \to 0$ as $n \to \infty$. Then*

$\hat{\gamma}_n(k) \xrightarrow{\mathbb{P}} \gamma$. *If, in addition,* $\sqrt{k}\,A(n/k) \to \lambda \in \mathbb{R}$, *then*

$$\sqrt{k}\,(\hat{\gamma}_n(k) - \gamma) \;\xrightarrow{d}\; \mathcal{N}\Big(\frac{\lambda}{1-\rho},\, \gamma^2\Big).$$

The choice of k controls the usual bias–variance trade-off; here it is treated as fixed by the data analyst.

2.2.2 Weissman extrapolation to extreme quantiles

The *Weissman extrapolation* (Weissman 1978) extrapolates from the intermediate-order statistic $X_{n-k:n}$ — which estimates the quantile $Q_X(1-k/n)$ at the intermediate level $1-k/n$ — to a quantile at a very-extreme level $1-p$ with $p = p_n \to 0$ as $n \to \infty$. Plugging the Hill estimator into the first-order identity $U(ty)/U(t) \to y^\gamma$ gives the Weissman estimator

$$\widehat{q}_n^{\star}(1-p \mid k) \;=\; \Big(\frac{k}{np}\Big)^{\hat{\gamma}_n(k)} X_{n-k\,:\,n}. \tag{2.2}$$

The theorem below, the Hill-estimator specialisation of Haan and Ferreira (2006, Theorem 4.3.8), covers the far-tail regime $np = o(k)$; the case $np = O(1)$ is very-extreme in the sense that only finitely many observations are expected beyond the target.

Theorem 2.6. *Assume* $\mathcal{C}_2(\gamma,\rho,A)$ *with* $\gamma > 0$ *and* $\rho < 0$. *Let* $k = k_n \to \infty$, $k/n \to 0$, $p = p_n \to 0$ *with* $k/(np) \to \infty$ *and* $\sqrt{k}/\log(k/(np)) \to \infty$. *Suppose* $\sqrt{k}\,A(n/k) \to \lambda \in \mathbb{R}$. *Then*

$$\frac{\sqrt{k}}{\log(k/(np))}\left(\frac{\widehat{q}_n^{\star}(1-p \mid k)}{Q_X(1-p)} - 1\right) \;\xrightarrow{d}\; \mathcal{N}\Big(\frac{\lambda}{1-\rho},\, \gamma^2\Big).$$

The log-denominator is the cost of extrapolation. Up to that factor the limit is the Hill limit, which is why Daouia, Padoan, et al. (2024, §2.3) pool Weissman estimators geometrically. This pooled-Weissman law is the source result against which the pooled extreme-expectile estimator in Chapter 3 will be compared.

2.3 Expectiles

Expectiles were introduced by Newey and Powell (1987) as asymmetric-least-squares analogues of quantiles. Their role as coherent and elicitable risk measures is developed in Bellini et al. (2014), Ziegel (2016), Gneiting (2011), and Ehm et al. (2016).

2.3.1 Definition

Let X be a real-valued random variable with $\mathbb{E}\,|X| < \infty$, and fix a level $\tau \in (0,1)$. The τ-*expectile* of X is defined as

$$\xi_\tau(X) \;:=\; \operatorname*{arg\,min}_{\theta \in \mathbb{R}}\; \mathbb{E}[\,\eta_\tau(X-\theta) - \eta_\tau(X)\,], \qquad \eta_\tau(u) := |\tau - \mathbb{1}\{u \le 0\}|\; u^2. \tag{2.3}$$

The subtraction of $\eta_\tau(X)$ makes the criterion finite under $\mathbb{E}\,|X| < \infty$ and does not affect the minimiser; at $\tau = 1/2$ one recovers the mean.

The convex objective (2.3) has a unique minimiser, and its first-order condition reduces to the implicit equation

$$\tau\;\mathbb{E}\big[(X - \xi_\tau(X))_+\big] \;=\; (1-\tau)\;\mathbb{E}\big[(\xi_\tau(X) - X)_+\big], \tag{2.4}$$

where $u_+ = \max(u, 0)$. Thus expectiles balance upper and lower first partial moments, unlike quantiles, which balance tail probabilities.

2.3.2 Basic properties

The following properties are immediate from (2.3) or (2.4); we collect them without proof and refer to Newey and Powell (1987, §2) and Bellini et al. (2014, Sections 2–3) for details.

Proposition 2.7. *Let X and Y be real-valued random variables with $\mathbb{E}\,|X|, \mathbb{E}\,|Y| < \infty$. For each $\tau \in (0,1)$:*

(i) ***Existence and uniqueness.*** *$\xi_\tau(X)$ exists and is uniquely defined.*

(ii) ***Law invariance.*** *If X and Y have the same distribution, then $\xi_\tau(X) = \xi_\tau(Y)$.*

(iii) ***Mean recovery.*** *$\xi_{1/2}(X) = \mathbb{E}\,X$.*

(iv) ***Location equivariance.*** *$\xi_\tau(X + c) = \xi_\tau(X) + c$ for $c \in \mathbb{R}$.*

(v) ***Positive homogeneity.*** *$\xi_\tau(\lambda X) = \lambda\,\xi_\tau(X)$ for $\lambda \geq 0$.*

(vi) ***Sign reflection.*** *$\xi_\tau(-X) = -\xi_{1-\tau}(X)$.*

(vii) ***Monotonicity in τ.*** *The map $\tau \mapsto \xi_\tau(X)$ is nondecreasing on $(0,1)$, and is strictly increasing when X is nondegenerate, with the boundary limits $\xi_\tau(X) \to \operatorname{ess\,inf}(X)$ as $\tau \to 0$ and $\xi_\tau(X) \to \operatorname{ess\,sup}(X)$ as $\tau \to 1$ (infinite limits allowed).*

(viii) ***Monotonicity in the argument.*** *If $X \leq Y$ almost surely, then $\xi_\tau(X) \leq \xi_\tau(Y)$.*

For losses, these properties give translation equivariance, positive homogeneity and monotonicity. If Y denotes a financial position and $\rho_\tau(Y) = \xi_\tau(-Y)$ is the associated capital requirement, then ρ_τ is coherent exactly when $\tau \geq 1/2$ (Bellini et al. 2014). Expectiles are also elicitable on classes with finite first moment; the asymmetric squared score is strictly consistent under finite second moments, and the general scoring class has Bregman-type and Choquet-type representations (Gneiting 2011; Ehm et al. 2016). These facts motivate high-level expectiles as tail-risk functionals, while the asymptotic theory below uses only (2.4) and the extreme expectile–quantile equivalence.

2.4 The high-expectile/high-quantile bridge

Chapter 3 uses expectile theory through the population relationship between a high expectile and the quantile of the same level. The stochastic extreme-quantile part is supplied by the pooled Weissman estimator of Daouia, Padoan, et al. (2024, §2.3), recalled in Section 2.5; the expectile-specific work is to control the deterministic bridge between ξ_τ and $Q_X(\tau)$. This section therefore records only the bridge statements needed later.

2.4.1 The expectile–quantile asymptotic equivalence

The single most important fact for extreme expectile inference is that, in the heavy-tailed regime, the expectile is asymptotically proportional to the quantile of the same level.

Proposition 2.8 (Asymptotic equivalence; Bellini et al. (2014), Daouia, Girard, et al. (2019), Corollary 1 with $p = 2$)**.** *Suppose Condition 2.4 holds. Then*

$$\frac{\xi_\tau(X)}{Q_X(\tau)} \xrightarrow{\tau\uparrow 1} \psi(\gamma) \;:=\; (\gamma^{-1} - 1)^{-\gamma}. \tag{2.5}$$

The bridge requires only the finite-mean range $\gamma < 1$ and lower-tail integrability; stronger restrictions used for direct empirical expectile estimation are not part of the quantile-based bridge route. The function ψ is continuous on $(0, 1)$, with $\psi(\gamma) \to 1$ as $\gamma \downarrow 0$ and $\psi(\gamma) \to \infty$ as $\gamma \uparrow 1$. It is *not* monotone: its logarithmic derivative

$$m(\gamma) \;:=\; \frac{d}{d\gamma} \log \psi(\gamma) \;=\; \frac{1}{1-\gamma} - \log\left(\gamma^{-1} - 1\right) \tag{2.6}$$

changes sign at $\gamma^\star \approx 0.2178$, where ψ reaches its minimum. In the pooled extreme-expectile theorem below, $m(\gamma)$ appears in the Taylor expansion of the bridge-Hill term; when $m(\gamma) = 0$, that term is even smaller at first order.

Under the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1)$ and $\mathbb{E}\, X^- < \infty$, the first-order equivalence (2.5) sharpens to the second-order expansion (Daouia, Girard, et al. 2020, Proposition 1(i))

$$\begin{aligned} \frac{\xi_\tau(X)}{Q_X(\tau)} &= \psi(\gamma)\Big[1 + c(\gamma,\rho)\, A((1-\tau)^{-1}) + \gamma(\gamma^{-1}-1)^{\gamma} \frac{\mathbb{E}\, X}{Q_X(\tau)} \\ &\qquad + o(|A((1-\tau)^{-1})|) + o(Q_X(\tau)^{-1})\Big], \\ c(\gamma,\rho) &:= \frac{(\gamma^{-1}-1)^{-\rho}}{1-\gamma-\rho} + \frac{(\gamma^{-1}-1)^{-\rho} - 1}{\rho}. \end{aligned} \tag{2.7}$$

At $\rho = 0$ the second term in $c(\gamma, \rho)$ is read by continuity, so $c(\gamma, 0) = m(\gamma)$. The A-term is the second-order expectile–quantile bias used later to control the population bridge term. The displayed first-moment term is $O(Q_X(\tau)^{-1})$; in Chapter 3 it is controlled on the pooled-Weissman scale by the explicit bridge-rate condition.

2.4.2 Relation to earlier expectile estimators

The iid extreme-expectile literature also studies direct asymmetric-least-squares estimators, quantile-based intermediate plug-in estimators, and Weissman-type expectile extrapolations from an intermediate expectile anchor (Daouia, Girard, et al. 2018; Daouia, Girard, et al. 2020; Davison et al. 2023). These results provide context for the use of the expectile–quantile bridge.

The estimator analysed in Chapter 3 performs the extreme extrapolation in the pooled-quantile component and then applies the population bridge at the target level. Consequently, the later argument relies on the first- and second-order bridge statements recorded in this section.

2.5 The optimal pooling framework

We now recall the pooling framework of Daouia, Padoan, et al. (2024). This framework provides the source theory for pooled tail-index and extreme-quantile inference before the expectile–quantile bridge is applied. The framework pools tail-index and extreme-quantile estimators from m samples, allowing heterogeneous sizes, second-order properties and cross-sample tail dependence. This general framework is recalled to identify the source objects and notation. The main theorem in Chapter 3 later uses only the independent common-distribution distributed special case.

2.5.1 Setup

Throughout Section 2.5, m is fixed while $n = \sum_{j=1}^m n_j \to \infty$. Let $\boldsymbol{X}_i = (X_{i,1}, \ldots, X_{i,m})^\top$, $i \geq 1$, be iid copies of $\boldsymbol{X} = (X_1, \ldots, X_m)^\top$. Margin j is observed for $i = 1, \ldots, n_j$ and has continuous right-heavy-tailed distribution F_j, tail quantile function $U_j = (1/\overline{F}_j)^\leftarrow$ and extreme-value index $\gamma_j > 0$; its order statistics are denoted $X_{1:n_j,j} \leq \cdots \leq X_{n_j:n_j,j}$. We write $Q_j(\tau) = F_j^\leftarrow(\tau)$, so that $U_j(t) = Q_j(1 - 1/t)$ for $t > 1$.

Tail inference in sample j uses its top $k_j + 1$ order statistics, where the effective sample size $k_j = k_j(n)$ satisfies $k_j \to \infty$ and $k_j/n_j \to 0$ as $n \to \infty$. The total effective sample size is $k = \sum_{j=1}^m k_j$. Assume the asymptotic-proportionality conditions

$$\frac{n_1}{n_j} \to b_j \in (0, \infty), \qquad \frac{k_1}{k_j} \to c_j \in (0, \infty), \tag{2.8}$$

with $b_1 = c_1 = 1$. The marginal F_j satisfies $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ (Condition 2.3). The thesis works under the *common-tail-index* assumption

$$(H_\gamma) \qquad \gamma_1 \;=\; \cdots \;=\; \gamma_m \;=\; \gamma.$$

Under this assumption, all common-γ vector limits below are expressed on the aggregate scale $\sqrt{k}$, with the constants c_j translating the local $\sqrt{k_j}$ rates.

The cross-sample asymptotic dependence is captured by a pairwise *tail-copula* structure (Daouia, Padoan, et al. 2024, §2.1).

Condition $\mathcal{J}(\boldsymbol{R})$ (tail-copula condition). For each pair (j, ℓ) with $j \neq \ell$, there exists a function $R_{j,\ell} : [0,\infty]^2 \setminus \{(\infty,\infty)\} \to [0,\infty)$ such that, for every $(x_j, x_\ell) \in [0,\infty]^2 \setminus \{(\infty,\infty)\}$,

$$\lim_{s\to\infty} s\, \mathbb{P}\big(1 - F_j(X_j) \leq x_j/s,\; 1 - F_\ell(X_\ell) \leq x_\ell/s\big) \;=\; R_{j,\ell}(x_j, x_\ell).$$

The case $R_{j,\ell} \equiv 0$ is asymptotic tail independence, implied by the independent samples used later in the thesis.

2.5.2 The pooled Hill estimator

Each sample contributes a Hill estimator $\hat{\gamma}_j(k_j)$ as in (2.1); we collect these into the random vector

$$\hat{\boldsymbol{\gamma}}_n \;=\; \big(\hat{\gamma}_1(k_1), \ldots, \hat{\gamma}_m(k_m)\big)^\top.$$

Given a weight vector $\boldsymbol{\omega} \in \mathbb{R}^m$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$, the *pooled Hill estimator* is the weighted arithmetic combination

$$\hat{\gamma}_n(\boldsymbol{\omega}) \;\coloneqq\; \boldsymbol{\omega}^\top \hat{\boldsymbol{\gamma}}_n \;=\; \sum_{j=1}^{m} \omega_j \, \hat{\gamma}_j(k_j). \tag{2.9}$$

The affine constraint $\boldsymbol{\omega}^\top \mathbf{1} = 1$ ensures consistency for the common γ under (H_γ).

The fundamental asymptotic result is the joint CLT for the marginal Hill estimators.

Theorem 2.9 (Daouia, Padoan, et al. (2024), Theorem 1)**.** *Assume $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ for every j, the tail-copula condition $\mathcal{J}(\boldsymbol{R})$, the proportionality conditions (2.8), and $\sqrt{k_j}\, A_j(n_j/k_j) \to \lambda_j \in \mathbb{R}$ for every j. Then*

$$\big(\sqrt{k_1}(\hat{\gamma}_1(k_1) - \gamma_1), \ldots, \sqrt{k_m}(\hat{\gamma}_m(k_m) - \gamma_m)\big)^\top \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{B}, \boldsymbol{V}),$$

with $\boldsymbol{B} = \big(\lambda_j/(1-\rho_j)\big)_{j=1}^m$ and $\boldsymbol{V}$ the symmetric matrix whose entries are

$$V_{j,\ell} \;=\; \begin{cases} \gamma_j^2, & j = \ell, \\ \dfrac{\gamma_j \gamma_\ell}{\max(b_j, b_\ell)\sqrt{c_j c_\ell}}\, R_{j,\ell}(b_j c_\ell,\, b_\ell c_j), & j < \ell. \end{cases}$$

The matrix is symmetric, so $V_{\ell,j} = V_{j,\ell}$ for $\ell > j$. Suppose, in addition, the common-γ condition (H_γ). Then the same marginal estimators, now viewed on the common $\sqrt{k}$ scale, satisfy

$$\sqrt{k}\,(\hat{\boldsymbol{\gamma}}_n - \gamma \mathbf{1}) \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{B}_{\boldsymbol{c}}, \boldsymbol{V}_{\boldsymbol{c}}),$$

where

$$(\boldsymbol{B}_{\boldsymbol{c}})_j = \sqrt{c_j \sum_{i=1}^{m} c_i^{-1}}\, \frac{\lambda_j}{1-\rho_j} \qquad j = 1, \ldots, m.$$

The covariance matrix $\boldsymbol{V}_{\boldsymbol{c}}$ is given by

$$(\boldsymbol{V}_{\boldsymbol{c}})_{j,\ell} = \left(\sum_{i=1}^{m} c_i^{-1}\right) \begin{cases} \gamma^2 c_j, & j = \ell, \\ \dfrac{\gamma^2}{\max(b_j, b_\ell)}\, R_{j,\ell}(b_j c_\ell, b_\ell c_j), & j < \ell, \end{cases}$$

with $(\boldsymbol{V}_{\boldsymbol{c}})_{\ell,j} = (\boldsymbol{V}_{\boldsymbol{c}})_{j,\ell}$ for $\ell > j$. Consequently, for any $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$,

$$\sqrt{k}\,(\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma) \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{\omega}^\top \boldsymbol{B}_{\boldsymbol{c}},\; \boldsymbol{\omega}^\top \boldsymbol{V}_{\boldsymbol{c}} \boldsymbol{\omega}).$$

For independent samples, all off-diagonal entries vanish and $\boldsymbol{V_c}$ becomes the diagonal matrix

$$\boldsymbol{V_c} \;=\; \gamma^2\left(\sum_{i=1}^{m} c_i^{-1}\right)\operatorname{diag}(c_1,\dots,c_m). \tag{2.10}$$

Equation (2.10) is the covariance form to be used when the main theorem is specialised to independent samples.

2.5.3 Variance- and AMSE-optimal weights

The relevant weights solve quadratic programmes under the affine constraint $\boldsymbol{\omega}^\top\mathbf{1} = 1$; unless stated otherwise, they need not be nonnegative. In Chapter 3, these formulas are applied only to the pooled-Weissman weights; they do not define or optimise the separate bridge-Hill weights introduced there.

Proposition 2.10 (Optimal weights; Daouia, Padoan, et al. (2024), Theorem 1; estimated-weight implementation in §2.2 and Corollary 1)**.** *Suppose $\boldsymbol{V_c}$ is positive definite. Under the common-γ condition (H_γ):*

(i) The variance-optimal weights, minimising $\boldsymbol{\omega}^\top\boldsymbol{V_c}\boldsymbol{\omega}$ subject to $\boldsymbol{\omega}^\top\mathbf{1} = 1$, are

$$\boldsymbol{\omega}^{\mathrm{Var}} \;=\; \frac{\boldsymbol{V_c}^{-1}\mathbf{1}}{\mathbf{1}^\top\boldsymbol{V_c}^{-1}\mathbf{1}},$$

with corresponding asymptotic variance $\boldsymbol{\omega}^{\mathrm{Var},\top}\boldsymbol{V_c}\boldsymbol{\omega}^{\mathrm{Var}} = (\mathbf{1}^\top\boldsymbol{V_c}^{-1}\mathbf{1})^{-1}$.

(ii) The AMSE-optimal weights, minimising $(\boldsymbol{\omega}^\top\boldsymbol{B_c})^2 + \boldsymbol{\omega}^\top\boldsymbol{V_c}\boldsymbol{\omega}$ subject to $\boldsymbol{\omega}^\top\mathbf{1} = 1$, are

$$\boldsymbol{\omega}^{\mathrm{AMSE}} \;=\; \frac{(1+\boldsymbol{B_c}^\top\boldsymbol{V_c}^{-1}\boldsymbol{B_c})\,\boldsymbol{V_c}^{-1}\mathbf{1} - (\mathbf{1}^\top\boldsymbol{V_c}^{-1}\boldsymbol{B_c})\,\boldsymbol{V_c}^{-1}\boldsymbol{B_c}}{(1+\boldsymbol{B_c}^\top\boldsymbol{V_c}^{-1}\boldsymbol{B_c})(\mathbf{1}^\top\boldsymbol{V_c}^{-1}\mathbf{1}) - (\mathbf{1}^\top\boldsymbol{V_c}^{-1}\boldsymbol{B_c})^2}.$$

In the independent-samples case, $\boldsymbol{V_c}$ is diagonal (Equation (2.10)) and the variance-optimal weights reduce to

$$\omega_j^{\mathrm{Var}} \;=\; \left(\sum_{i=1}^{m} c_i^{-1}\right)^{-1} c_j^{-1} \;\sim\; \frac{k_j}{k}, \tag{2.11}$$

where the equivalence uses $c_i^{-1} = \lim k_i/k_1$. The finite-sample distributed version is

$$\tilde{\boldsymbol{\omega}}_n^{\mathrm{Var}} = (k_1/k,\dots,k_m/k)^\top,$$

which is exactly normalised and converges to $\boldsymbol{\omega}^{\mathrm{Var}}$.

Feasible weights replace $\boldsymbol{V_c}$ and $\boldsymbol{B_c}$ by consistent plug-ins (Daouia, Padoan, et al. 2024, §2.2). If $\hat{\boldsymbol{\omega}}^\top\mathbf{1} = 1$ exactly and $\hat{\boldsymbol{\omega}} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$, then $\hat{\gamma}_n(\hat{\boldsymbol{\omega}})$ is $\sqrt{k}$-equivalent to $\hat{\gamma}_n(\boldsymbol{\omega})$; the difficult practical step is stable estimation of $\boldsymbol{V_c}$ and the bias quantities.

2.5.4 Weighted geometric pooling of extreme quantiles

For estimation of an extreme quantile $q(1-p)$ with $p = p(n) \to 0$, each sample contributes a Weissman estimator

$$\hat{q}_j^\star(1-p \mid k_j) \;=\; \left(\frac{k_j}{n_j\, p}\right)^{\hat{\gamma}_j(k_j)} X_{n_j-k_j:n_j,\,j}.$$

Because Weissman extrapolation is multiplicative, pooling is geometric on the original scale, equivalently linear on the log scale. It requires asymptotic equivalence of the marginal extreme quantiles:

Condition 2.11 (Tail homoskedasticity ($\mathcal{H}$))**.** For every $j \neq \ell$, $U_j(t)/U_\ell(t) \to 1$ as $t \to \infty$.

Equivalently, $Q_j(1-p)/Q_\ell(1-p) \to 1$ as $p \downarrow 0$. The pooled extreme-quantile estimator of Daouia, Padoan, et al. (2024, §2.3) is

$$\widehat{q}_n^\star(1-p \mid \boldsymbol{\omega}) \;\coloneqq\; \prod_{j=1}^{m} \left[\hat{q}_j^\star(1-p \mid k_j)\right]^{\omega_j}. \tag{2.12}$$

Taking logs reduces pooling to weighted sums of the local identities $\log \hat{q}_j^\star(1-p \mid k_j) = \log(k_j/(n_j p))\, \hat{\gamma}_j(k_j) + \log X_{n_j-k_j:n_j,\,j}$; so the Hill part is handled by Theorem 2.9; (2.8) gives $\log(k_j/(n_j p)) = \log(k/(np)) + O(1)$.

Theorem 2.12 (Daouia, Padoan, et al. (2024), Theorem 2)**.** *Assume the hypotheses of Theorem 2.9, the common-γ condition (H_γ), the tail-homoskedasticity condition ($\mathcal{H}$), and, in addition, $\rho_j < 0$ for every $j = 1, \dots, m$. Let $p = p(n) \to 0$ with $k/(np) \to \infty$ and $\sqrt{k}/\log(k/(np)) \to \infty$. Then for any $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$ and any $j \in \{1, \dots, m\}$,*

$$\frac{\sqrt{k}}{\log(k/(np))}\left(\frac{\widehat{q}_n^\star(1-p \mid \boldsymbol{\omega})}{Q_j(1-p)} - 1\right) \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{\omega}^\top \boldsymbol{B_c},\; \boldsymbol{\omega}^\top \boldsymbol{V_c}\boldsymbol{\omega}).$$

The strict $\rho_j < 0$ assumption is part of the Weissman theorem. Together with the bias-rate assumptions $\sqrt{k_j}\, A_j(n_j/k_j) \to \lambda_j$ in Theorem 2.9, it also supplies the rate needed to recenter by any marginal $Q_j(1-p)$ under ($\mathcal{H}$) (Daouia, Padoan, et al. 2024, Supplement, Lemma B.2): for any j, ℓ,

$$\log Q_j(1-p) - \log Q_\ell(1-p) = O(\,|A_j(1/p)|\,) + O(\,|A_\ell(1/p)|\,) = o(k^{-1/2}),$$

along the Weissman sequence.

Thus the pooled Weissman and pooled Hill limits share the same mean, variance and optimal weights up to the log-rate factor. This shared structure is the reason Chapter 3 can use the weight criteria of Daouia, Padoan, et al. (2024) for the pooled-Weissman component.

2.5.5 Estimated weights and plug-in objects

The optimal weights above are oracle quantities because they depend on the asymptotic bias and covariance objects. For inference they are replaced either by deterministic allocation weights, such as $(k_1/k, \ldots, k_m/k)^\top$ in the independent distributed case, or by estimated weights based on plug-in estimates of the same objects. The following result records the stability property used in Chapter 3. Estimated weights are used there only in the pooled-Weissman role; the bridge-Hill vector remains deterministic.

Proposition 2.13 (Estimated pooling weights; Daouia, Padoan, et al. (2024), Supplement, Theorem A.1; Theorem 2 and Corollary 8)**.** *Assume the common-γ conditions of Theorem 2.9. Let $\widehat{\boldsymbol{\omega}}_n$ be a random weight vector satisfying*

$$\widehat{\boldsymbol{\omega}}_n^\top \mathbf{1} = 1, \qquad \widehat{\boldsymbol{\omega}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\omega},$$

for some deterministic $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$. Then

$$\sqrt{k}\,(\hat{\gamma}_n(\widehat{\boldsymbol{\omega}}_n) - \gamma) \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}^\top \boldsymbol{B_c}, \boldsymbol{\omega}^\top \boldsymbol{V_c}\boldsymbol{\omega}).$$

If, in addition, the assumptions of Theorem 2.12 hold, then

$$\frac{\sqrt{k}}{\log(k/(np))}\left(\frac{\widehat{q}_n^\star(1-p \mid \widehat{\boldsymbol{\omega}}_n)}{Q_j(1-p)} - 1\right) \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}^\top \boldsymbol{B_c}, \boldsymbol{\omega}^\top \boldsymbol{V_c}\boldsymbol{\omega}), \qquad j = 1, \ldots, m.$$

In the independent common-distribution distributed case, the same statement holds with the diagonal covariance matrix (2.10) and the distributed bias constants of Daouia, Padoan, et al. (2024, Corollary 5).

The supplement's Theorem A.1 is the generic estimated-weight transfer: exact affine normalisation and convergence in probability of the weights make the random-weight pooled estimator asymptotically equivalent to the pooled estimator with limiting deterministic weights. Applied to Theorem 2.9, this gives the Hill statement. The Weissman statement is the corresponding random-weight version of Theorem 2.12; in the distributed common-marginal case it is the published Corollary 8 of Daouia, Padoan, et al. (2024).

The next proposition separates the probabilistic consistency of the estimated bias and variance objects from the algebra of applying a chosen weight vector.

Proposition 2.14 (Plug-in bias and variance)**.** *Let $\widehat{\boldsymbol{B}}_{\boldsymbol{c}n}$ and $\widehat{\boldsymbol{V}}_{\boldsymbol{c}n}$ be estimators of $\boldsymbol{B_c}$ and $\boldsymbol{V_c}$ satisfying*

$$\widehat{\boldsymbol{B}}_{\boldsymbol{c}n} \xrightarrow{\mathbb{P}} \boldsymbol{B_c}, \qquad \widehat{\boldsymbol{V}}_{\boldsymbol{c}n} \xrightarrow{\mathbb{P}} \boldsymbol{V_c},$$

and let $\widehat{\boldsymbol{\omega}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$. Define

$$\widehat{B}_{\widehat{\omega},n} = \widehat{\boldsymbol{\omega}}_n^\top \widehat{\boldsymbol{B}}_{\boldsymbol{c}n}, \qquad \widehat{V}_{\widehat{\omega},n} = \widehat{\boldsymbol{\omega}}_n^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}n} \widehat{\boldsymbol{\omega}}_n.$$

Then

$$\widehat{B}_{\widehat{\omega},n} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}^\top \boldsymbol{B_c}, \qquad \widehat{V}_{\widehat{\omega},n} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}^\top \boldsymbol{V_c}\boldsymbol{\omega}.$$

All vectors and matrices have fixed dimension. The maps $(w, b) \mapsto w^\top b$ and $(w, V) \mapsto w^\top V w$ are continuous, so the result follows from the continuous mapping theorem.

Daouia, Padoan, et al. (2024, §2.2 and Corollary 1) give one source route to the consistency assumptions in Proposition 2.14 through estimated second-order and tail-dependence quantities. That route is not assumption-free: in the source it is stated under $\rho_j < 0$, the parametric second-order form $A_j(t) = \gamma\beta_j t^{\rho_j}$, positive definiteness of the asymptotic covariance matrix, consistency of the β_j estimators, and $(\hat{\rho}_j - \rho_j)\log n_j = o_P(1)$. In the common-distribution distributed case, their Corollaries 6–7 give the corresponding deterministic variance allocation and plug-in AMSE constructions under the analogous common second-order parameter consistency assumptions.

The log-scale interval for the expectile estimator in Chapter 3 is derived from the transferred plug-in centring and variance objects after the expectile decomposition.

2.5.6 Distributed inference as a special case

In the independent common-distribution distributed setting, the samples are iid subdivisions of one common population. Then:

- the common-γ condition (H_γ) and the tail-homoskedasticity condition $(\mathcal{H})$ are trivially satisfied;
- the tail copula $R_{j,\ell}$ between any pair of distinct samples vanishes identically (independence implies tail independence);
- $\boldsymbol{V_c}$ in Theorem 2.9 becomes the diagonal matrix (2.10);
- the variance-optimal weights of Proposition 2.10 reduce to $\omega_j^{\mathrm{Var}} \propto k_j$ as in (2.11).

Under the stronger balanced-allocation assumptions studied by Daouia, Padoan, et al. (2024, Corollary 6, Theorem 3 and Corollary 8), the variance-optimally pooled Hill and pooled Weissman estimators match the corresponding unfeasible estimators built from the entire combined sample at first order. This oracle equivalence is useful benchmark context for distributed inference. The main theorem in Chapter 3 uses the more direct consequence needed for the expectile problem: in the independent common-distribution case, the pooled-Weissman bias and variance objects are the diagonal distributed objects displayed above.

Chapter 3

Pooled extreme expectiles

This chapter develops the pooled extreme-expectile estimator studied in the thesis. We work in the finite-m distributed iid setting of Section 2.5.6: each machine contains observations from the same continuous heavy-tailed loss distribution, and the target is the common marginal expectile ξ_{τ_n}. In this setting the population target is unambiguous, and the pooled extreme-quantile component can be handled through the distributed pooled-Weissman theorem of Daouia, Padoan, et al. (2024, Corollary 8).

The construction combines the geometrically pooled Weissman estimator with the asymptotic bridge between high expectiles and high quantiles. The role of the chapter is to show, from an exact log decomposition, which terms remain visible on the pooled-Weissman scale. Under the bridge-rate condition imposed below, the first-order law is inherited from the pooled extreme-quantile component. Accordingly, first-order weight optimisation concerns the Daouia–Padoan–Stupfler weights in the pooled-Weissman part.

3.1 Estimator and exact decomposition

Let $p_n = 1 - \tau_n$, $n = \sum_{j=1}^m n_j$, $k = \sum_{j=1}^m k_j$, and

$$\ell_n = \log\left(\frac{k}{np_n}\right).$$

For each machine j, let $\hat{\gamma}_j(k_j)$ be the local Hill estimator and

$$\hat{q}_j^\star(1 - p_n \mid k_j) = \left(\frac{k_j}{n_j p_n}\right)^{\hat{\gamma}_j(k_j)} X_{n_j - k_j : n_j, j}$$

be the local Weissman estimator. For an affine weight vector $\boldsymbol{\omega}^\top \mathbf{1} = 1$, the geometrically pooled Weissman estimator is

$$\widehat{q}_n^\star(1 - p_n \mid \boldsymbol{\omega}) = \prod_{j=1}^m \left[\hat{q}_j^\star(1 - p_n \mid k_j)\right]^{\omega_j}.$$

For another affine weight vector $\boldsymbol{\nu}^\top \mathbf{1} = 1$, define the pooled Hill estimator used in the outer bridge by

$$\hat\gamma_n(\boldsymbol{\nu}) = \sum_{j=1}^{m} \nu_j \hat\gamma_j(k_j).$$

The two-weight pooled extreme-expectile estimator is

$$\widehat{\xi}^{\text{pool},\star}_{\tau_n}(\boldsymbol{\nu},\boldsymbol{\omega}) = \psi(\hat\gamma_n(\boldsymbol{\nu}))\,\widehat{q}^{\star}_n(1-p_n \mid \boldsymbol{\omega}), \qquad \psi(\gamma) = (\gamma^{-1}-1)^{-\gamma}. \tag{3.1}$$

The vector $\boldsymbol{\omega}$ weights the pooled-Weissman quantile estimator. The vector $\boldsymbol{\nu}$ weights only the Hill estimator appearing in the outer expectile bridge. The diagonal construction $\boldsymbol{\nu} = \boldsymbol{\omega}$ is a special case, not a first-order necessity.

Lemma 3.1 (Exact log decomposition). *Let $Q(1-p_n)$ denote the common marginal quantile and let ξ_{τ_n} denote the corresponding expectile. Then*

$$\log \frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n}(\boldsymbol{\nu},\boldsymbol{\omega})}{\xi_{\tau_n}} = A_n(\boldsymbol{\omega}) + B_n(\boldsymbol{\nu}) - C_n, \tag{3.2}$$

where

$$A_n(\boldsymbol{\omega}) = \log \frac{\widehat{q}^{\star}_n(1-p_n \mid \boldsymbol{\omega})}{Q(1-p_n)}, \qquad B_n(\boldsymbol{\nu}) = \log \psi(\hat\gamma_n(\boldsymbol{\nu})) - \log \psi(\gamma),$$

and

$$C_n = \log \frac{\xi_{\tau_n}}{\psi(\gamma) Q(1-p_n)}.$$

Proof. By the definition of the estimator,

$$\begin{aligned}
\log \frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n}(\boldsymbol{\nu},\boldsymbol{\omega})}{\xi_{\tau_n}} &= \log \frac{\psi(\hat\gamma_n(\boldsymbol{\nu}))\widehat{q}^{\star}_n(1-p_n \mid \boldsymbol{\omega})}{\xi_{\tau_n}} \\
&= \log \psi(\hat\gamma_n(\boldsymbol{\nu})) + \log \widehat{q}^{\star}_n(1-p_n \mid \boldsymbol{\omega}) - \log \xi_{\tau_n}.
\end{aligned}$$

Insert and subtract the two deterministic quantities $\log Q(1-p_n)$ and $\log \psi(\gamma)$. This gives

$$\begin{aligned}
\log \frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n}(\boldsymbol{\nu},\boldsymbol{\omega})}{\xi_{\tau_n}} &= \{\log \widehat{q}^{\star}_n(1-p_n \mid \boldsymbol{\omega}) - \log Q(1-p_n)\} \\
&\quad + \{\log \psi(\hat\gamma_n(\boldsymbol{\nu})) - \log \psi(\gamma)\} \\
&\quad + \{\log \psi(\gamma) + \log Q(1-p_n) - \log \xi_{\tau_n}\}.
\end{aligned}$$

The first bracket is $A_n(\boldsymbol{\omega})$, the second bracket is $B_n(\boldsymbol{\nu})$, and the third bracket equals

$$\log \frac{\psi(\gamma) Q(1-p_n)}{\xi_{\tau_n}} = -\log \frac{\xi_{\tau_n}}{\psi(\gamma) Q(1-p_n)} = -C_n.$$

Combining the three brackets yields (3.2). □

3.2 A deterministic two-weight theorem

The following theorem gives the first-order limit under a rate condition that makes the deterministic population gap between expectiles and quantiles negligible on the pooled-Weissman scale.

Theorem 3.2 (Pooled extreme expectile, deterministic weights)**.** *Let m be fixed. Suppose the observations across and within machines are iid copies of a common continuous loss variable X. Let $U(t) = Q(1 - 1/t)$ be the common tail quantile, and assume $\mathcal{C}_2(\gamma, \rho, A)$ with $0 < \gamma < 1$ and $\rho < 0$. Assume $\mathbb{E}\, X^- < \infty$ and $U(t) > 0$ for all sufficiently large t.*

Let $n = \sum_{j=1}^m n_j$, $k = \sum_{j=1}^m k_j$, and assume

$$\frac{n_1}{n_j} \to b_j \in (0, \infty), \qquad \frac{k_1}{k_j} \to c_j \in (0, \infty),$$

with $b_1 = c_1 = 1$. Suppose

$$k \to \infty, \qquad \frac{k}{n} \to 0, \qquad \sqrt{k}\, A(n/k) \to \lambda \in \mathbb{R}.$$

Let $p_n = 1 - \tau_n$, $\ell_n = \log\{k/(np_n)\}$, and assume the extreme Weissman regime

$$p_n \to 0, \qquad \frac{k}{np_n} \to \infty, \qquad \frac{\sqrt{k}}{\ell_n} \to \infty.$$

Finally impose the expectile-bridge rate condition

$$\eta_n = \frac{\sqrt{k}}{\ell_n Q(1 - p_n)} \to 0. \tag{3.3}$$

Let $\boldsymbol{\nu}_n, \boldsymbol{\omega}_n \in \mathbb{R}^m$ be deterministic weight sequences satisfying exact affine normalisation and finite-dimensional convergence,

$$\boldsymbol{\nu}_n^\top \mathbf{1} = 1, \qquad \boldsymbol{\omega}_n^\top \mathbf{1} = 1, \qquad \boldsymbol{\nu}_n \to \boldsymbol{\nu}, \qquad \boldsymbol{\omega}_n \to \boldsymbol{\omega},$$

for finite vectors $\boldsymbol{\nu}, \boldsymbol{\omega} \in \mathbb{R}^m$. Fixed deterministic weights are included by taking constant sequences. No non-negativity is imposed unless stated separately. Then the estimator (3.1) satisfies

$$\frac{\sqrt{k}}{\ell_n} \log \frac{\widehat{\xi}^{\,\mathrm{pool},\star}_{\tau_n}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}} \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}), \tag{3.4}$$

where

$$B_{\boldsymbol{\omega}} = \frac{\lambda}{1 - \rho} \sum_{j=1}^m d_j^\rho \omega_j, \qquad V_{\boldsymbol{\omega}} = \gamma^2 \left(\sum_{j=1}^m c_j^{-1} \right) \left(\sum_{j=1}^m c_j \omega_j^2 \right),$$

and

$$d_j = \frac{c_j}{b_j} \frac{\sum_{i=1}^m c_i^{-1}}{\sum_{i=1}^m b_i^{-1}}.$$

Moreover,

$$\frac{\sqrt{k}}{\ell_n}\left(\frac{\widehat{\xi}^{\mathrm{pool},\star}_{\tau_n}(\boldsymbol{\nu}_n,\boldsymbol{\omega}_n)}{\xi_{\tau_n}}-1\right) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}},V_{\boldsymbol{\omega}}). \tag{3.5}$$

Proof. Start from the exact identity (3.2). We treat the three terms separately on the scale $s_n=\sqrt{k}/\ell_n$.

For $A_n(\boldsymbol{\omega}_n)$, use the random-weight version of the common-marginal pooled-Weissman theorem recalled in Proposition 2.13, applied to the exactly normalised deterministic sequence $\boldsymbol{\omega}_n\to\boldsymbol{\omega}$. It gives

$$\frac{\sqrt{k}}{\ell_n}\left(\frac{\widehat{q}^{\star}_n(1-p_n\mid\boldsymbol{\omega}_n)}{Q(1-p_n)}-1\right) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}},V_{\boldsymbol{\omega}}).$$

Set

$$x_n=\frac{\widehat{q}^{\star}_n(1-p_n\mid\boldsymbol{\omega}_n)}{Q(1-p_n)}-1.$$

The condition $\sqrt{k}/\ell_n\to\infty$ implies $x_n=O_P(\ell_n/\sqrt{k})=o_P(1)$. Hence

$$\log(1+x_n)=x_n+O_P(x_n^2)$$

and

$$\frac{\sqrt{k}}{\ell_n}\{\log(1+x_n)-x_n\}=O_P\!\left(\frac{\ell_n}{\sqrt{k}}\right)=o_P(1).$$

Therefore

$$\frac{\sqrt{k}}{\ell_n}A_n(\boldsymbol{\omega}_n) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}},V_{\boldsymbol{\omega}}). \tag{3.6}$$

For $B_n(\boldsymbol{\nu}_n)$, write

$$g(x)=\log\psi(x)=-x\log(x^{-1}-1).$$

On every compact subinterval of $(0,1)$, g is twice continuously differentiable, with

$$g'(x)=m(x)=\frac{1}{1-x}-\log(x^{-1}-1),\qquad g''(x)=\frac{1}{(1-x)^2}+\frac{1}{1-x}+\frac{1}{x}.$$

The aggregate assumptions imply the local bias calibration required by the vector pooled Hill CLT. Indeed, $k_j\to\infty$, $k_j/n_j\to 0$, and

$$\frac{k_j}{k}\to\frac{c_j^{-1}}{\sum_{i=1}^m c_i^{-1}},\qquad \frac{n_j}{n}\to\frac{b_j^{-1}}{\sum_{i=1}^m b_i^{-1}}.$$

Consequently,

$$\frac{n_j/k_j}{n/k}\to d_j=\frac{c_j}{b_j}\,\frac{\sum_{i=1}^m c_i^{-1}}{\sum_{i=1}^m b_i^{-1}},$$

and regular variation of A gives

$$\sqrt{k_j}\,A(n_j/k_j)=\sqrt{\frac{k_j}{k}}\,\sqrt{k}\,A(n/k)\,\frac{A(n_j/k_j)}{A(n/k)}\to\lambda\left(\frac{c_j^{-1}}{\sum_{i=1}^m c_i^{-1}}\right)^{1/2} d_j^{\rho}.$$

Thus the common-marginal distributed assumptions of Theorem 3.2 meet the local assumptions of Theorem 2.9. In particular,

$$\sqrt{k}\{\hat{\boldsymbol{\gamma}}_n - \gamma\mathbf{1}\} = O_P(1).$$

Since $\boldsymbol{\nu}_n \to \boldsymbol{\nu}$, the sequence $(\boldsymbol{\nu}_n)$ is bounded, and

$$\sqrt{k}\{\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma\} = \boldsymbol{\nu}_n^\top \sqrt{k}\{\hat{\boldsymbol{\gamma}}_n - \gamma\mathbf{1}\} = O_P(1).$$

Thus $\hat{\gamma}_n(\boldsymbol{\nu}_n) \xrightarrow{\mathbb{P}} \gamma$. Hence, with probability tending to one, the line segment between $\hat{\gamma}_n(\boldsymbol{\nu}_n)$ and γ lies in a fixed compact subset of $(0,1)$. Taylor's theorem on that event gives

$$B_n(\boldsymbol{\nu}_n) = m(\gamma)\{\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma\} + O_P\Big(\{\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma\}^2\Big) = m(\gamma)\{\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma\} + O_P(k^{-1}).$$

Consequently

$$\frac{\sqrt{k}}{\ell_n} B_n(\boldsymbol{\nu}_n) = \frac{m(\gamma)}{\ell_n}\sqrt{k}\{\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma\} + O_P\left(\frac{1}{\ell_n\sqrt{k}}\right) = o_P(1), \tag{3.7}$$

because $\ell_n \to \infty$, which follows from $k/(np_n) \to \infty$. If $m(\gamma) = 0$, the displayed conclusion follows from the quadratic remainder alone.

It remains to control C_n. Put

$$a_n = A(1/p_n), \qquad b_n = Q(1-p_n)^{-1}, \qquad \alpha_\gamma = \gamma(\gamma^{-1} - 1)^\gamma, \qquad \mu = \mathbb{E}\, X.$$

The second-order expectile–quantile expansion (2.7) gives, after dividing by $\psi(\gamma)$,

$$\Delta_n \coloneqq \frac{\xi_{\tau_n}}{\psi(\gamma)Q(1-p_n)} - 1 = c(\gamma,\rho)a_n + \alpha_\gamma \mu b_n + o(|a_n|) + o(b_n).$$

Since $\Delta_n \to 0$,

$$C_n = \log(1 + \Delta_n) = \Delta_n + O(\Delta_n^2).$$

The regular variation of A, the strict inequality $\rho < 0$, and

$$\frac{1/p_n}{n/k} = \frac{k}{np_n} \to \infty$$

imply

$$\frac{A(1/p_n)}{A(n/k)} = \left(\frac{k}{np_n}\right)^\rho \{1 + o(1)\} \to 0.$$

Together with $\sqrt{k}A(n/k) \to \lambda$, this yields

$$\frac{\sqrt{k}}{\ell_n} a_n \to 0.$$

The imposed bridge-rate condition (3.3) is exactly

$$\frac{\sqrt{k}}{\ell_n} b_n \to 0.$$

Therefore all linear terms in C_n are $o(1)$ after multiplication by $\sqrt{k}/\ell_n$. The logarithmic square remainder is also negligible, since

$$\frac{\sqrt{k}}{\ell_n}(|a_n| + b_n)^2 \to 0$$

follows from $(\sqrt{k}/\ell_n)|a_n| \to 0$, $(\sqrt{k}/\ell_n)b_n \to 0$, and $|a_n| + b_n \to 0$. Hence

$$\frac{\sqrt{k}}{\ell_n} C_n \to 0. \tag{3.8}$$

Combining (3.6), (3.7), and (3.8) in the exact decomposition gives (3.4) by Slutsky's theorem.

For the relative-error statement, define

$$L_n = \log \frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}}.$$

The log limit implies $L_n = O_P(\ell_n/\sqrt{k}) = o_P(1)$. Therefore

$$e^{L_n} - 1 = L_n + O_P(L_n^2)$$

and

$$\frac{\sqrt{k}}{\ell_n}\{e^{L_n} - 1 - L_n\} = O_P\left(\frac{\ell_n}{\sqrt{k}}\right) = o_P(1).$$

This proves (3.5). □

Remark 3.3 (What the theorem does and does not identify). The limit in Theorem 3.2 depends on $\boldsymbol{\omega}$, the limiting geometric pooled-Weissman weights, but not on $\boldsymbol{\nu}$, the limiting bridge-Hill weights. Thus $\boldsymbol{\nu}$ is first-order unidentified in the present regime. The diagonal estimator $\boldsymbol{\nu}_n = \boldsymbol{\omega}_n$ is a permissible deterministic convention when the common deterministic sequence satisfies the theorem's convergence condition, and the practical convention used for plug-in inference below is

$$\boldsymbol{\nu}_n^k = \left(\frac{k_1}{k}, \dots, \frac{k_m}{k}\right)^\top.$$

Neither convention is an optimality result for $\boldsymbol{\nu}$. Any optimal choice of bridge-Hill weights would require a lower-order theory not developed here.

3.3 Weight consequences

The preceding theorem determines the relevant first-order weight criterion. The leading bias and variance coincide with those of the Daouia–Padoan–Stupfler pooled-Weissman component in the distributed common-marginal case. Define

$$\boldsymbol{B}_{\boldsymbol{c}}^{\text{dist}} = \frac{\lambda}{1-\rho}(d_1^\rho, \dots, d_m^\rho)^\top, \qquad \boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}} = \gamma^2\left(\sum_{i=1}^m c_i^{-1}\right)\operatorname{diag}(c_1, \dots, c_m).$$

Then $B_{\boldsymbol{\omega}} = \boldsymbol{\omega}^\top \boldsymbol{B}_{\boldsymbol{c}}^{\text{dist}}$ and $V_{\boldsymbol{\omega}} = \boldsymbol{\omega}^\top \boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}} \boldsymbol{\omega}$.

Corollary 3.4 (First-order criteria for $\boldsymbol{\omega}$)**.** *Under the assumptions of Theorem 3.2, the first-order variance criterion for the pooled extreme-expectile estimator is*

$$\boldsymbol{\omega}^\top \boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}} \boldsymbol{\omega},$$

and the first-order AMSE criterion is

$$(\boldsymbol{\omega}^\top \boldsymbol{B}_{\boldsymbol{c}}^{\text{dist}})^2 + \boldsymbol{\omega}^\top \boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}} \boldsymbol{\omega},$$

both under the affine constraint $\boldsymbol{\omega}^\top \mathbf{1} = 1$. Hence the deterministic/oracle variance- and AMSE-optimal formulas of Daouia, Padoan, et al. (2024) apply to $\boldsymbol{\omega}$. These criteria concern $\boldsymbol{\omega}$ only; the bridge-Hill weights $\boldsymbol{\nu}$ do not enter because $B_n(\boldsymbol{\nu}_n)$ is lower order under (3.3). In particular,

$$\boldsymbol{\omega}^{\text{Var}} = \frac{(\boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}})^{-1} \mathbf{1}}{\mathbf{1}^\top (\boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}})^{-1} \mathbf{1}},$$

so, in the independent distributed case,

$$\omega_j^{\text{Var}} = \frac{c_j^{-1}}{\sum_{i=1}^m c_i^{-1}}, \qquad j = 1, \dots, m.$$

If $\boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}}$ is positive definite, the AMSE-optimal weights are

$$\boldsymbol{\omega}^{\text{AMSE}} = \frac{(1 + (\boldsymbol{B}_{\boldsymbol{c}}^{\text{dist}})^\top (\boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}})^{-1} \boldsymbol{B}_{\boldsymbol{c}}^{\text{dist}}) (\boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}})^{-1} \mathbf{1} - (\mathbf{1}^\top (\boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}})^{-1} \boldsymbol{B}_{\boldsymbol{c}}^{\text{dist}}) (\boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}})^{-1} \boldsymbol{B}_{\boldsymbol{c}}^{\text{dist}}}{(1 + (\boldsymbol{B}_{\boldsymbol{c}}^{\text{dist}})^\top (\boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}})^{-1} \boldsymbol{B}_{\boldsymbol{c}}^{\text{dist}}) (\mathbf{1}^\top (\boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}})^{-1} \mathbf{1}) - (\mathbf{1}^\top (\boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}})^{-1} \boldsymbol{B}_{\boldsymbol{c}}^{\text{dist}})^2}.$$

No first-order variance or AMSE criterion for $\boldsymbol{\nu}$ is identified by Theorem 3.2.

Proof. By Theorem 3.2,

$$\frac{\sqrt{k}}{\ell_n} \log \frac{\widehat{\xi}_{\tau_n}^{\text{pool},\star}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}} \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}).$$

Thus the scaled first-order mean-square criterion is $B_{\boldsymbol{\omega}}^2 + V_{\boldsymbol{\omega}}$, and the unscaled criterion differs only by the common factor ℓ_n^2/k. The same criterion applies to relative error by (3.5). Substituting $B_{\boldsymbol{\omega}} = \boldsymbol{\omega}^\top \boldsymbol{B}_{\boldsymbol{c}}^{\text{dist}}$ and $V_{\boldsymbol{\omega}} = \boldsymbol{\omega}^\top \boldsymbol{V}_{\boldsymbol{c}}^{\text{dist}} \boldsymbol{\omega}$ gives exactly the quadratic criteria recalled in Proposition 2.10. The variance and AMSE formulas follow from that result. □

3.4 Estimated pooled-Weissman weights and inference

The deterministic theorem is the base result. Estimated weights are introduced only in the pooled-Weissman role $\boldsymbol{\omega}$, where Daouia, Padoan, et al. (2024) already provide random-weight quantile limits under their own consistency conditions. The bridge-Hill vector remains deterministic in this chapter. When the estimated weights or plug-in bias and variance objects are taken from the Daouia–Padoan–Stupfler constructions, the source-side consistency assumptions recalled in

Section 2.5.5 are imposed.

The only new source input is the random-weight version of the Daouia–Padoan–Stupfler pooled-Weissman limit: exact affine normalisation $\widehat{\boldsymbol{\omega}}_n^\top \mathbf{1} = 1$ and convergence $\widehat{\boldsymbol{\omega}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$ give the same first-order quantile limit as the deterministic vector $\boldsymbol{\omega}$. The rest of the argument is a transfer through the exact decomposition.

Corollary 3.5 (Estimated-$\boldsymbol{\omega}$ transfer)**.** *Assume the conditions of Theorem 3.2, with the deterministic pooled-Weissman weight sequence in the estimator replaced by a random $\widehat{\boldsymbol{\omega}}_n$ satisfying the estimated-weight conditions of Proposition 2.13, with limiting vector $\boldsymbol{\omega}$. Let $\boldsymbol{\nu}_n$ be any deterministic bridge-Hill weight sequence satisfying the exact affine normalisation and convergence conditions of Theorem 3.2; in particular one may take $\boldsymbol{\nu}_n = \boldsymbol{\nu}_n^k = (k_1/k, \ldots, k_m/k)^\top$. Then*

$$\frac{\sqrt{k}}{\ell_n} \log \frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n}(\boldsymbol{\nu}_n, \widehat{\boldsymbol{\omega}}_n)}{\xi_{\tau_n}} \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}).$$

The same limit holds on relative-error scale.

Proof. The exact decomposition becomes

$$\log \frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n}(\boldsymbol{\nu}_n, \widehat{\boldsymbol{\omega}}_n)}{\xi_{\tau_n}} = A_n(\widehat{\boldsymbol{\omega}}_n) + B_n(\boldsymbol{\nu}_n) - C_n.$$

Proposition 2.13 gives the random-weight relative-error limit for the pooled-Weissman component. With

$$x_n(\widehat{\boldsymbol{\omega}}_n) = \frac{\widehat{q}^\star_n(1 - p_n \mid \widehat{\boldsymbol{\omega}}_n)}{Q(1 - p_n)} - 1,$$

this gives

$$\frac{\sqrt{k}}{\ell_n} x_n(\widehat{\boldsymbol{\omega}}_n) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}).$$

Since $\sqrt{k}/\ell_n \to \infty$, $x_n(\widehat{\boldsymbol{\omega}}_n) = O_P(\ell_n/\sqrt{k}) = o_P(1)$, and the relative-to-log Taylor step used in the proof of Theorem 3.2 gives

$$\frac{\sqrt{k}}{\ell_n} A_n(\widehat{\boldsymbol{\omega}}_n) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}).$$

The bridge-Hill term is unchanged because $\boldsymbol{\nu}_n$ is deterministic, exactly normalised, and convergent. The proof of (3.7) therefore applies verbatim. The population bridge term C_n is deterministic and is still controlled by (3.8). Slutsky's theorem gives the log limit, and the final exponential Taylor step in Theorem 3.2 gives the relative-error version. □

Thus any source-admissible Daouia–Padoan–Stupfler construction of $\widehat{\boldsymbol{\omega}}_n$ transfers in the pooled-Weissman role, including estimated variance- and AMSE-optimal weights when their source assumptions give $\widehat{\boldsymbol{\omega}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$. This statement concerns $\boldsymbol{\omega}$ only. It does not optimise $\boldsymbol{\nu}$, and it does not cover a random bridge-Hill convention such as $\boldsymbol{\nu} = \widehat{\boldsymbol{\omega}}_n$.

Corollary 3.6 (Plug-in studentisation and log-scale interval)**.** *Let*

$$\widehat{\xi}^{\mathrm{plug}}_{\tau_n} = \widehat{\xi}^{\mathrm{pool},\star}_{\tau_n}(\boldsymbol{\nu}^k_n, \widehat{\boldsymbol{\omega}}_n), \qquad \boldsymbol{\nu}^k_n = \left(\frac{k_1}{k}, \ldots, \frac{k_m}{k}\right)^{\top}.$$

Under the assumptions of Corollary 3.5, let $\widehat{\boldsymbol{B}_{\boldsymbol{c}n}}^{\mathrm{dist}}$ *and* $\widehat{\boldsymbol{V}_{\boldsymbol{c}n}}^{\mathrm{dist}}$ *be plug-in estimators satisfying*

$$\widehat{\boldsymbol{B}_{\boldsymbol{c}n}}^{\mathrm{dist}} \xrightarrow{\mathbb{P}} \boldsymbol{B}^{\mathrm{dist}}_{\boldsymbol{c}}, \qquad \widehat{\boldsymbol{V}_{\boldsymbol{c}n}}^{\mathrm{dist}} \xrightarrow{\mathbb{P}} \boldsymbol{V}^{\mathrm{dist}}_{\boldsymbol{c}}.$$

These consistency assumptions belong to the selected Daouia–Padoan–Stupfler plug-in route; they are not consequences of the expectile decomposition alone. No expectile-specific bias or variance estimator is introduced here. Define

$$\widehat{B}_{\widehat{\boldsymbol{\omega}},n} = \widehat{\boldsymbol{\omega}}^{\top}_n \widehat{\boldsymbol{B}_{\boldsymbol{c}n}}^{\mathrm{dist}}, \qquad \widehat{V}_{\widehat{\boldsymbol{\omega}},n} = \widehat{\boldsymbol{\omega}}^{\top}_n \widehat{\boldsymbol{V}_{\boldsymbol{c}n}}^{\mathrm{dist}} \widehat{\boldsymbol{\omega}}_n,$$

and assume $V_{\boldsymbol{\omega}} > 0$*. Then*

$$T^{\mathrm{plug}}_n = \frac{\frac{\sqrt{k}}{\ell_n} \log\left(\frac{\widehat{\xi}^{\mathrm{plug}}_{\tau_n}}{\xi_{\tau_n}}\right) - \widehat{B}_{\widehat{\boldsymbol{\omega}},n}}{\sqrt{\widehat{V}_{\widehat{\boldsymbol{\omega}},n}}} \xrightarrow{d} \mathcal{N}(0,1).$$

Let $z_{1-\alpha/2}$ *be the standard normal quantile. On the high-probability event that the local Weissman estimators are positive and* $\widehat{V}_{\widehat{\boldsymbol{\omega}},n} > 0$*, define*

$$I^{\mathrm{plug}}_{n,1-\alpha} = \Bigg[\widehat{\xi}^{\mathrm{plug}}_{\tau_n} \exp\left\{-\frac{\ell_n}{\sqrt{k}}\left(\widehat{B}_{\widehat{\boldsymbol{\omega}},n} + z_{1-\alpha/2}\sqrt{\widehat{V}_{\widehat{\boldsymbol{\omega}},n}}\right)\right\},$$
$$\widehat{\xi}^{\mathrm{plug}}_{\tau_n} \exp\left\{-\frac{\ell_n}{\sqrt{k}}\left(\widehat{B}_{\widehat{\boldsymbol{\omega}},n} - z_{1-\alpha/2}\sqrt{\widehat{V}_{\widehat{\boldsymbol{\omega}},n}}\right)\right\}\Bigg].$$

Then

$$\mathbb{P}\left\{\xi_{\tau_n} \in I^{\mathrm{plug}}_{n,1-\alpha}\right\} \to 1-\alpha.$$

Proof. Corollary 3.5 gives

$$\frac{\sqrt{k}}{\ell_n} \log\left(\frac{\widehat{\xi}^{\mathrm{plug}}_{\tau_n}}{\xi_{\tau_n}}\right) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}).$$

Proposition 2.14, applied with $\boldsymbol{B}_{\boldsymbol{c}} = \boldsymbol{B}^{\mathrm{dist}}_{\boldsymbol{c}}$ and $\boldsymbol{V}_{\boldsymbol{c}} = \boldsymbol{V}^{\mathrm{dist}}_{\boldsymbol{c}}$, gives

$$\widehat{B}_{\widehat{\boldsymbol{\omega}},n} \xrightarrow{\mathbb{P}} B_{\boldsymbol{\omega}}, \qquad \widehat{V}_{\widehat{\boldsymbol{\omega}},n} \xrightarrow{\mathbb{P}} V_{\boldsymbol{\omega}}.$$

The additional expectile terms have already been absorbed in Corollary 3.5, so the centring and variance objects remain the pooled-Weissman objects supplied by the source theory. Together with $V_{\boldsymbol{\omega}} > 0$, the displayed convergences give the standard normal limit by Slutsky's theorem. Since $\widehat{V}_{\widehat{\boldsymbol{\omega}},n} \xrightarrow{\mathbb{P}} V_{\boldsymbol{\omega}} > 0$, the event $\widehat{V}_{\widehat{\boldsymbol{\omega}},n} > 0$ has probability tending to one. Eventual upper-tail positivity also gives positivity of the high order statistics, and hence of all local Weissman estimators, with probability tending to one. Solving

$$\left| T_n^{\text{plug}} \right| \leq z_{1-\alpha/2}$$

for the unknown ξ_{τ_n} gives the displayed multiplicative interval, so its asymptotic coverage follows from the studentised limit. □

Remark 3.7 (Scope of the inference corollary)*.* The interval in Corollary 3.6 is obtained by inverting the plug-in-centred expectile statistic above. It is therefore a consequence of the transferred expectile limit, rather than a direct restatement of the quantile interval in Daouia, Padoan, et al. (2024). The centering and variance objects remain those of the pooled-Weissman component because, under $\eta_n \to 0$, the additional expectile-bridge terms are lower order on that scale. The corollary is consequently conditional on the selected Daouia–Padoan–Stupfler construction providing consistent plug-in objects for $B_{\boldsymbol{\omega}}$ and $V_{\boldsymbol{\omega}}$; it does not make an additional claim about simulations, finite-sample calibration, or optimal bridge-Hill weights.

Chapter 4

Finite-sample study

This chapter reports a finite-sample study for the estimator developed in Chapter 3. The purpose is deliberately narrow. The simulations study

$$\widehat{\xi}_{\tau}^{\mathrm{pool},\star}(\boldsymbol{\nu}_n^k,\boldsymbol{\omega}) = \psi(\hat{\gamma}_n(\boldsymbol{\nu}_n^k))\widehat{q}_n^{\star}(1-p \mid \boldsymbol{\omega}), \qquad \boldsymbol{\nu}_n^k = (k_1/k,\dots,k_m/k)^{\top},$$

where $p = 1-\tau$ and $k = \sum_{j=1}^{m} k_j$. Thus the bridge-Hill weights are fixed at the deterministic threshold-allocation convention, while the pooled-Weissman weights $\boldsymbol{\omega}$ vary across the Daouia–Padoan–Stupfler choices studied below. The role of $\boldsymbol{\nu}_n^k$ is that of a transparent implementation convention rather than an optimality claim; all weight variation in the study concerns $\boldsymbol{\omega}$.

The empirical conclusions should be read as diagnostics for the promoted first-order theory, not as new mathematical claims. In the representative imbalanced design, the Daouia–Padoan–Stupfler variance weights make the distributed estimator track the centralised benchmark closely for point estimation. Equal pooled-Weissman weights are less efficient under imbalance. In the main equal-threshold grid, AMSE weights are almost indistinguishable from variance weights; in the separate heterogeneous-threshold diagnostic they become more informative, but can also produce unstable signed weights. The log-scale plug-in intervals under-cover in several models, so the finite-sample evidence supports the estimator more strongly than it supports the nominal first-order interval calibration.

4.1 Estimators and implementation

For each machine j, the local Hill estimator and Weissman quantile estimator are computed using the `HTailIndex` and `extQuantile` routines from the `ExtremeRisks` package (Padoan 2026). The simulation code adds only the thesis-specific pooling layer: geometric pooling of local Weissman estimates, the map ψ, the deterministic bridge-Hill convention $\boldsymbol{\nu}_n^k$, and the plug-in summary calculations.

The following estimators are recorded:

- *equal*: $\omega_j = 1/m$;

- *Daouia–Padoan–Stupfler variance*: $\omega_j = k_j/k$, the independent distributed version of the variance weights in this common-tail design. Since the variance-optimal formula reduces to this deterministic threshold allocation, the oracle and plug-in variance-weight versions coincide;
- *Daouia–Padoan–Stupfler AMSE oracle*: the AMSE formula evaluated with the population second-order parameters of the data-generating model;
- *Daouia–Padoan–Stupfler AMSE plug-in*: the same formula with second-order inputs estimated using `evt0` (Belzile et al. 2026);
- *centralised*: the corresponding single-sample estimator computed from the combined sample, used only as a benchmark.

All distributed estimators use the same bridge-Hill vector $\boldsymbol{\nu}_n^k$. The plug-in AMSE estimator randomises only $\boldsymbol{\omega}$. For exact Pareto models, the second-order bias coefficient is zero in the design, so the plug-in AMSE route is treated as source-ineligible rather than as a distinct estimated AMSE procedure. The oracle AMSE row then collapses to the same point estimator as the variance rule. For the centralised benchmark interval, the centring and variance objects are the corresponding single-sample oracle objects obtained by taking $m = 1$.

For each estimator with available centring and variance objects, inference is assessed with the log-scale interval from Corollary 3.6. In one Monte Carlo replication, if $\widehat{B}_\omega$ and $\widehat{V}_\omega$ denote the selected Daouia–Padoan–Stupfler centring and variance objects, the interval is

$$\widehat{\xi}\left[\exp\left\{-\frac{\ell}{\sqrt{k}}(\widehat{B}_\omega + z_{0.975}\sqrt{\widehat{V}_\omega})\right\}, \exp\left\{-\frac{\ell}{\sqrt{k}}(\widehat{B}_\omega - z_{0.975}\sqrt{\widehat{V}_\omega})\right\}\right], \qquad \ell = \log\left(\frac{k}{np}\right).$$

The interval is included as a diagnostic of the first-order plug-in route, not as a claim of finite-sample calibration.

4.2 Design

The study stays within the iid common-marginal setting of the theorem. The data-generating models are Pareto with $\gamma = 0.25$, Pareto with $\gamma = 0.60$, Burr with $\gamma = 0.25$ and $\rho = -1$, Burr with $\gamma = 0.60$ and $\rho = -1$, and the absolute Student distribution with three degrees of freedom. The latter has $\gamma = 1/3$. The true expectile ξ_τ is computed numerically from the population expectile equation

$$\tau\ \mathbb{E}[(X - \xi_\tau)_+] = (1-\tau)\ \mathbb{E}[(\xi_\tau - X)_+],$$

rather than by the asymptotic expectile–quantile bridge.

The final finite-sample simulation uses 5000 Monte Carlo replications per scenario. The grid has 55 scenarios: 20 main-grid scenarios, 15 threshold-sensitivity scenarios, 15 target-sensitivity scenarios, and 5 heterogeneous-threshold scenarios. The main grid uses $n = 10000$, $k_j/n_j = 0.05$, $np = 5$, $m \in \{5, 10\}$, and either balanced allocation or a strong imbalanced allocation with machine proportions increasing geometrically from 1 to 10. The threshold block fixes $m = 10$

and strong allocation, then varies $k_j/n_j \in \{0.03, 0.05, 0.10\}$. The target block fixes the same design and varies $np \in \{1, 5, 10\}$.

The heterogeneous-threshold block is separate. It keeps $m = 10$, strong allocation, $n = 10000$, and $np = 5$, but assigns larger threshold fractions to smaller machines and smaller threshold fractions to larger machines. The profile is rescaled so the aggregate k/n remains close to 0.05. This block is included because the main grid uses a common local threshold fraction, in which case $(k_1/k, \dots, k_m/k)^\top$ and $(n_1/n, \dots, n_m/n)^\top$ coincide.

Performance is measured by the Monte Carlo mean of the log relative error, the root mean squared log relative error, empirical coverage of the nominal 95% log interval, average log interval width, invalid-estimate rate, and weight-stability diagnostics. The code also records the exact finite-sample decomposition terms

$$A_n(\boldsymbol{\omega}) = \log\{\widehat{q}_n^\star(1-p \mid \boldsymbol{\omega})/Q(1-p)\}, \qquad B_n(\boldsymbol{\nu}) = \log\psi\{\hat{\gamma}_n(\boldsymbol{\nu})\} - \log\psi(\gamma),$$

and

$$C_n = \log\{\xi_\tau/(\psi(\gamma)Q(1-p))\}.$$

These diagnostics connect the Monte Carlo evidence to Lemma 3.1; they are not used to estimate new asymptotic constants.

4.3 Point estimation and pooled-Weissman weights

Table 4.1 reports the representative main design with $m = 10$, strong sample-size imbalance, $k_j/n_j = 0.05$, and $np = 5$. The estimator using Daouia–Padoan–Stupfler variance weights is close to the centralised benchmark across the five models. Its log RMSE is 0.223 versus 0.226 for Burr heavy, 0.070 versus 0.069 for Burr light, 0.204 versus 0.204 for Pareto heavy, 0.090 versus 0.089 for Pareto light, and 0.142 versus 0.143 for the absolute Student model. This is the main positive finite-sample finding: in this grid, the promoted distributed estimator does not pay a large point-estimation penalty relative to the single-sample benchmark.

Equal pooled-Weissman weights are worse under strong imbalance. The effect is most visible in the heavier-tailed models, where the equal-weight log RMSE is 0.245 for Burr heavy and 0.231 for Pareto heavy, compared with 0.223 and 0.204 under Daouia–Padoan–Stupfler variance weights. The oracle and plug-in AMSE weights are essentially the same as the variance weights in this equal-threshold main design. This statement is deliberately local to the main grid; Table 4.9 gives AMSE weights a more discriminating environment by varying the local threshold fractions.

The blank plug-in AMSE rows for exact Pareto models are not invalid point-estimates. They mark the conservative source decision described above: the plug-in AMSE route is not reported as an active estimated second-order procedure when the model has zero second-order bias coefficient.

Table 4.1: Point-estimation performance in the representative main design ($m = 10$, strong allocation, $k_j/n_j = 0.05$, $np = 5$).

DGP	m	Regime	Estimator	Log bias	Log RMSE	Valid rate
Burr heavy	10	strong	Centralised	0.077	0.226	1.000
Burr heavy	10	strong	DPS AMSE oracle	0.075	0.223	1.000
Burr heavy	10	strong	DPS AMSE plug-in	0.075	0.223	1.000
Burr heavy	10	strong	DPS variance	0.075	0.223	1.000
Burr heavy	10	strong	Equal	0.070	0.245	1.000
Burr light	10	strong	Centralised	-0.039	0.069	1.000
Burr light	10	strong	DPS AMSE oracle	-0.041	0.070	1.000
Burr light	10	strong	DPS AMSE plug-in	-0.041	0.070	1.000
Burr light	10	strong	DPS variance	-0.041	0.070	1.000
Burr light	10	strong	Equal	-0.043	0.084	1.000
Pareto heavy	10	strong	Centralised	-0.007	0.204	1.000
Pareto heavy	10	strong	DPS AMSE oracle	-0.012	0.204	1.000
Pareto heavy	10	strong	DPS AMSE plug-in	–	–	0.000
Pareto heavy	10	strong	DPS variance	-0.012	0.204	1.000
Pareto heavy	10	strong	Equal	-0.012	0.231	1.000
Pareto light	10	strong	Centralised	-0.069	0.089	1.000
Pareto light	10	strong	DPS AMSE oracle	-0.071	0.090	1.000
Pareto light	10	strong	DPS AMSE plug-in	–	–	0.000
Pareto light	10	strong	DPS variance	-0.071	0.090	1.000
Pareto light	10	strong	Equal	-0.071	0.100	1.000
Abs. Student-3	10	strong	Centralised	0.110	0.143	1.000
Abs. Student-3	10	strong	DPS AMSE oracle	0.109	0.142	1.000
Abs. Student-3	10	strong	DPS AMSE plug-in	0.109	0.142	1.000
Abs. Student-3	10	strong	DPS variance	0.109	0.142	1.000
Abs. Student-3	10	strong	Equal	0.108	0.154	1.000

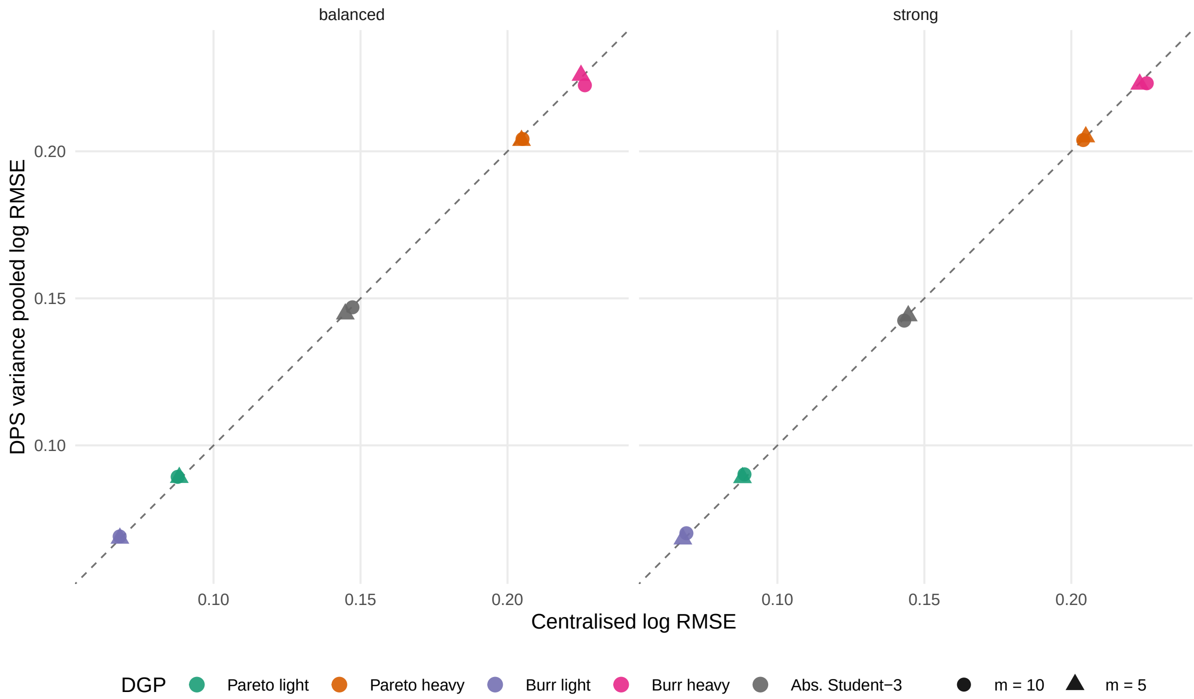

Figure 4.1: Log RMSE of the pooled estimator using Daouia–Padoan–Stupfler variance weights against the centralised benchmark in the main grid with $k_j/n_j = 0.05$ and $np = 5$. The dashed line is equality. Points near the line indicate that the pooled route tracks the centralised benchmark closely.

4.4 Deterministic bridge-Hill sensitivity

The main specification fixes the bridge-Hill weights at $\boldsymbol{\nu}_n^k = (k_1/k, \ldots, k_m/k)^\top$. To check whether this convention has visible finite-sample consequences, Table 4.2 repeats the representative main design for the Daouia–Padoan–Stupfler variance pooled-Weissman weights while replacing $\boldsymbol{\nu}_n^k$ by two other deterministic conventions: equal bridge-Hill weights and sample-allocation weights $(n_1/n, \ldots, n_m/n)^\top$. This is only a sensitivity check. It does not search for an optimal $\boldsymbol{\nu}$, and it does not use the random estimated $\boldsymbol{\omega}$ as a bridge-Hill vector.

In the representative main design, the threshold and sample-allocation conventions coincide because k_j/n_j is constant across machines. The table therefore mainly compares this allocation convention against equal bridge-Hill weights. Equal bridge-Hill weights have little effect in the light-tailed rows, but are less favourable in Burr heavy, Pareto heavy, and the absolute Student model. The conclusion is not that $\boldsymbol{\nu}_n^k$ is optimal. The conclusion is only that the deterministic threshold-allocation convention is not visibly harmful in this main grid.

Table 4.3 makes the comparison more informative by breaking the equality between threshold and sample-allocation weights. The differences remain modest. In Burr heavy, the sample bridge vector has the lowest log RMSE, 0.231, while $\boldsymbol{\nu}_n^k$ has slightly higher coverage, 0.744. In the absolute Student model, sample weights again have the lowest log RMSE, while $\boldsymbol{\nu}_n^k$ remains close. In the light-tailed models, all three bridge-Hill conventions are nearly indistinguishable. These patterns are consistent with the theorem's message: $\boldsymbol{\nu}$ is first-order unidentified under the chosen scaling, but lower-order finite-sample differences can still be visible.

Table 4.2: Sensitivity to deterministic bridge-Hill weights in the representative main design. The pooled-Weissman weights are fixed at the Daouia–Padoan–Stupfler variance choice.

DGP	Bridge weights	Log bias	Log RMSE	Coverage
Burr heavy	Equal	0.077	0.236	0.720
Burr heavy	Sample	0.075	0.223	0.751
Burr heavy	ν^k	0.075	0.223	0.751
Burr light	Equal	-0.041	0.070	0.593
Burr light	Sample	-0.041	0.070	0.591
Burr light	ν^k	-0.041	0.070	0.591
Pareto heavy	Equal	-0.007	0.216	–
Pareto heavy	Sample	-0.012	0.204	–
Pareto heavy	ν^k	-0.012	0.204	–
Pareto light	Equal	-0.071	0.090	–
Pareto light	Sample	-0.071	0.090	–
Pareto light	ν^k	-0.071	0.090	–
Abs. Student-3	Equal	0.110	0.144	0.910
Abs. Student-3	Sample	0.109	0.142	0.920
Abs. Student-3	ν^k	0.109	0.142	0.920

Table 4.3: Sensitivity to deterministic bridge-Hill weights under heterogeneous local threshold fractions. The aggregate k/n is kept close to 0.05, and the pooled-Weissman weights are fixed at the Daouia–Padoan–Stupfler variance choice.

DGP	Bridge weights	Log bias	Log RMSE	Coverage	RMS scaled B_n
Burr heavy	Equal	0.106	0.242	0.742	0.554
Burr heavy	Sample	0.090	0.231	0.733	0.504
Burr heavy	ν^k	0.097	0.234	0.744	0.501
Burr light	Equal	-0.034	0.067	0.603	0.023
Burr light	Sample	-0.035	0.067	0.599	0.020
Burr light	ν^k	-0.034	0.067	0.600	0.020
Pareto heavy	Equal	-0.008	0.212	–	0.415
Pareto heavy	Sample	-0.010	0.210	–	0.416
Pareto heavy	ν^k	-0.010	0.208	–	0.387
Pareto light	Equal	-0.071	0.091	–	0.015
Pareto light	Sample	-0.071	0.091	–	0.015
Pareto light	ν^k	-0.072	0.091	–	0.014
Abs. Student-3	Equal	0.138	0.167	0.918	0.275
Abs. Student-3	Sample	0.127	0.157	0.925	0.222
Abs. Student-3	ν^k	0.132	0.162	0.923	0.245

4.5 Plug-in interval performance

Table 4.4 reports nominal 95% log intervals in the representative main design. For non-Pareto distributed rows, the table reports the source-admissible plug-in route. For exact Pareto models, the distributed plug-in interval rows are omitted because the plug-in second-order route is source-ineligible; the centralised oracle benchmark is still shown. The same distinction is made visually in Figure 4.2: exact-Pareto distributed interval cells are marked by dashes rather than interpreted as zero coverage.

The intervals are not uniformly calibrated at this sample size. The distributed Burr heavy intervals cover about 75% of replications, the Burr light intervals cover about 59%, and the absolute Student intervals cover about 92%. The centralised oracle benchmark also under-covers in every model, with coverage between 0.708 and 0.770. Hence the problem is not simply that the sample has been split across machines. The finite sample is also feeling first-order centring, variance, and bridge approximations.

Figure 4.3 gives a targeted diagnostic for this undercoverage. It uses oracle centring and variance objects to compare coverage against both the exact population expectile and the first-order bridge target $\psi(\gamma)Q(1-p)$, for two diagnostic designs separate from the final main grid. In the light-tailed rows, much of the exact-target coverage loss disappears when the bridge target is used. For example, in the $n = 10000$, $k/n = 0.05$ diagnostic, Burr light moves from about 71% exact coverage to about 92% bridge-target coverage, and light Pareto moves from about 70% to about 95%. In the $n = 100000$, $k = n^{0.45}$ diagnostic, exact-target coverage for these light-tailed rows is also higher, about 89% for Burr light and 87% for light Pareto. This supports the interpretation that some of the observed undercoverage is a finite-sample bridge effect. The heavy-tailed rows are less improved by this diagnostic, so the conclusion remains cautionary rather than corrective.

Table 4.4: Empirical coverage and log-width of nominal 95% intervals in the representative main design.

DGP	m	Regime	Estimator	Coverage	Log width	CI rate	η_n	Bridge gap
Burr heavy	10	strong	Centralised	0.755	0.486	1.000	0.051	0.009
Burr heavy	10	strong	DPS AMSE plug-in	0.751	0.498	1.000	0.051	0.009
Burr heavy	10	strong	DPS variance	0.751	0.498	1.000	0.051	0.009
Burr light	10	strong	Centralised	0.718	0.202	1.000	0.724	0.058
Burr light	10	strong	DPS AMSE plug-in	0.591	0.208	1.000	0.724	0.058
Burr light	10	strong	DPS variance	0.591	0.208	1.000	0.724	0.058
Pareto heavy	10	strong	Centralised	0.770	0.486	1.000	0.051	0.012
Pareto light	10	strong	Centralised	0.708	0.202	1.000	0.724	0.070
Abs. Student-3	10	strong	Centralised	0.751	0.270	1.000	0.297	0.030
Abs. Student-3	10	strong	DPS AMSE plug-in	0.920	0.305	1.000	0.297	0.030
Abs. Student-3	10	strong	DPS variance	0.920	0.305	1.000	0.297	0.030

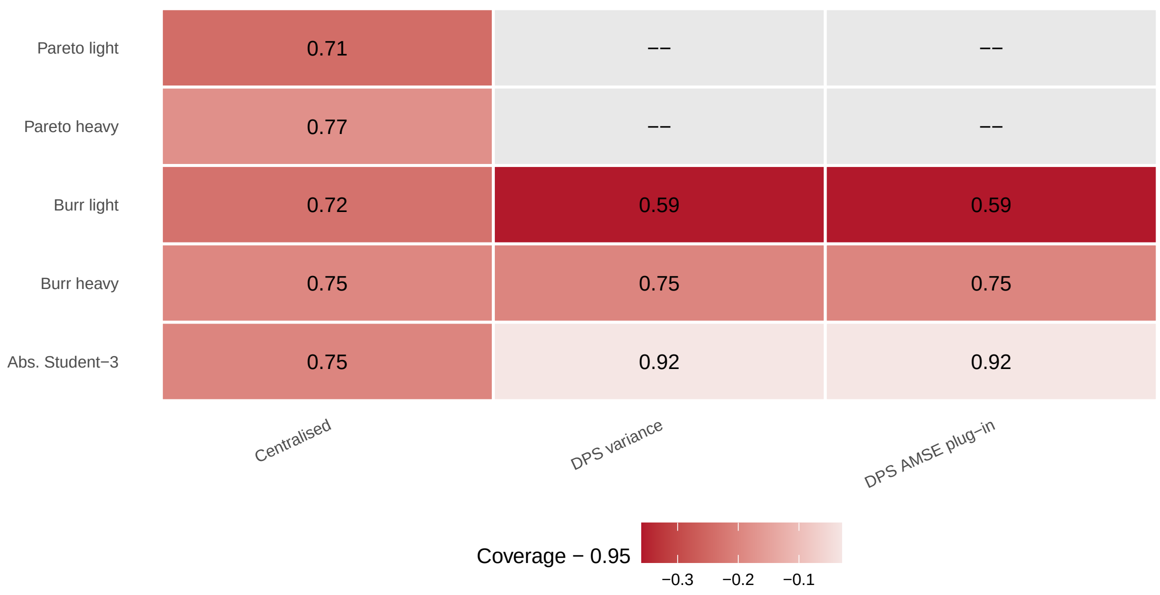

Figure 4.2: Empirical coverage in the representative main design ($m = 10$, strong allocation, $k_j/n_j = 0.05$, $np = 5$). Cell labels give coverage; colour records the deviation from the nominal 95% level. Cells marked by dashes are source-ineligible exact-Pareto distributed interval rows.

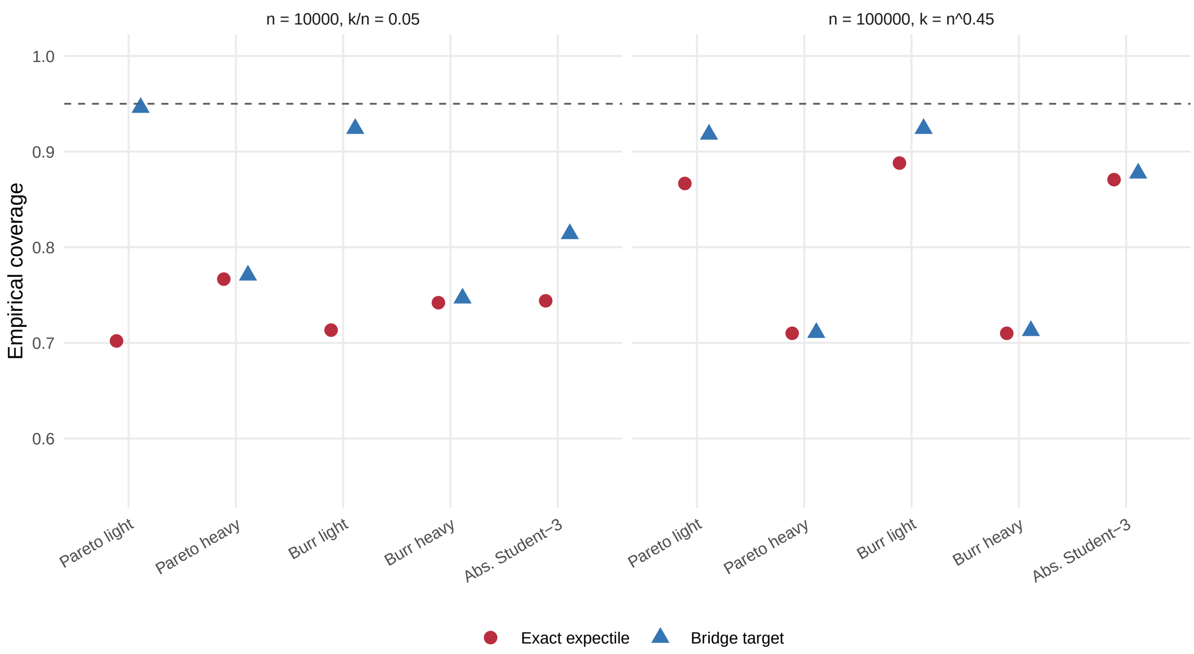

Figure 4.3: Interval diagnostic comparing empirical coverage against the exact population expectile and against the first-order bridge target $\psi(\gamma)Q(1-p)$. The diagnostic uses oracle centring and variance objects and is separate from the final main grid.

4.6 Decomposition diagnostics

Table 4.5 records the scaled decomposition terms in the representative main design for the estimator using Daouia–Padoan–Stupfler variance weights with $\boldsymbol{\nu} = \boldsymbol{\nu}_n^k$. The columns report root mean squared values of $(\sqrt{k}/\ell)A_n$, $(\sqrt{k}/\ell)B_n$, and $(\sqrt{k}/\ell)C_n$, together with the scaled log error. The final column is the numerical reconstruction check for

$$\log(\widehat{\xi}_\tau/\xi_\tau) = A_n(\boldsymbol{\omega}) + B_n(\boldsymbol{\nu}) - C_n.$$

The table answers the same question as the proof: which terms are visible at the pooled-Weissman scale? In the asymptotic theorem $A_n(\boldsymbol{\omega})$ carries the first-order law, while $B_n(\boldsymbol{\nu})$ and C_n vanish after multiplication by $\sqrt{k}/\ell$. At the simulated sample size, that separation is only partial. The pooled-Weissman component is the largest stochastic component in most rows, but the bridge-Hill term is not negligible in the heavier-tailed cases: its RMS scaled value is 0.474 in Burr heavy and 0.382 in Pareto heavy, compared with 0.638 and 0.610 for $A_n(\boldsymbol{\omega})$. In contrast, $B_n(\boldsymbol{\nu})$ is almost invisible in the light-tailed Pareto and Burr designs.

The population bridge term C_n shows the opposite pattern. It is small in the heavier-tailed Pareto and Burr cases, but it is comparable to, or larger than, the pooled-Weissman component in the light-tailed cases. This helps explain why point-estimation performance and interval calibration need not move together: some finite-sample error is coming from the population expectile–quantile bridge rather than from the distributed quantile pooling itself. The reconstruction check is at numerical roundoff, confirming that the diagnostics read the exact decomposition rather than an approximate post-hoc split.

Table 4.5: Finite-sample decomposition ledger in the representative main design. The estimator uses Daouia–Padoan–Stupfler variance pooled-Weissman weights and $\boldsymbol{\nu} = \boldsymbol{\nu}_n^k$.

DGP	RMS scaled log error	RMS scaled A_n	RMS scaled B_n	$\lvert$scaled $C_n\rvert$	Check
Burr heavy	1.080	0.638	0.474	0.045	0.00
Burr light	0.340	0.269	0.019	0.279	0.00
Pareto heavy	0.987	0.610	0.382	0.057	0.00
Pareto light	0.437	0.255	0.013	0.338	0.00
Abs. Student-3	0.690	0.589	0.222	0.143	0.00

4.7 Sensitivity to thresholds and targets

The threshold sensitivity in Table 4.6 shows that point error and interval calibration can move in different directions. Burr light is the clearest example: increasing k_j/n_j from 0.03 to 0.10 lowers the log RMSE of the Daouia–Padoan–Stupfler variance-weighted estimator from 0.084 to 0.047, while coverage falls from about 70% to about 32%. In Pareto heavy, where the exact Pareto model removes the Burr-type second-order bias, increasing the threshold improves the corresponding log RMSE from 0.248 to 0.156, but the distributed interval coverage entries are source-ineligible and therefore left blank. In the absolute Student case, the largest threshold worsens log RMSE from 0.116 to 0.285, even though coverage stays near 90%. Threshold choice therefore cannot be defended by a single point-error criterion.

Table 4.7 varies the target through np. Moving from $np = 1$ to $np = 10$ shortens the extrapolation distance, and this lowers log RMSE in Burr heavy and the absolute Student model. It does not uniformly improve calibration. Burr light coverage falls from about 79% at $np = 1$ to about 43% at $np = 10$, Burr heavy coverage falls from about 78% to about 70%, and absolute Student coverage falls from about 92% to about 88%. The target sensitivity again points to an interaction between second-order bias, threshold choice, and first-order interval centring, not to a simple monotone improvement when the target is made less extreme.

Table 4.6: Threshold sensitivity for $m = 10$, strong allocation and $np = 5$.

DGP	Threshold rule	Estimator	k/n	Log RMSE	Coverage
Burr heavy	fraction=0.03	DPS AMSE plug-in	0.029	0.255	0.731
Burr heavy	fraction=0.03	DPS variance	0.029	0.255	0.731
Burr heavy	fraction=0.05	DPS AMSE plug-in	0.050	0.228	0.732
Burr heavy	fraction=0.05	DPS variance	0.050	0.228	0.733
Burr heavy	fraction=0.1	DPS AMSE plug-in	0.100	0.260	0.744
Burr heavy	fraction=0.1	DPS variance	0.100	0.260	0.743
Burr light	fraction=0.03	DPS AMSE plug-in	0.029	0.084	0.695
Burr light	fraction=0.03	DPS variance	0.029	0.084	0.696
Burr light	fraction=0.05	DPS AMSE plug-in	0.050	0.069	0.580
Burr light	fraction=0.05	DPS variance	0.050	0.069	0.579
Burr light	fraction=0.1	DPS AMSE plug-in	0.100	0.047	0.314
Burr light	fraction=0.1	DPS variance	0.100	0.047	0.316
Pareto heavy	fraction=0.03	DPS AMSE plug-in	0.029	–	–
Pareto heavy	fraction=0.03	DPS variance	0.029	0.248	–
Pareto heavy	fraction=0.05	DPS AMSE plug-in	0.050	–	–
Pareto heavy	fraction=0.05	DPS variance	0.050	0.208	–
Pareto heavy	fraction=0.1	DPS AMSE plug-in	0.100	–	–
Pareto heavy	fraction=0.1	DPS variance	0.100	0.156	–
Pareto light	fraction=0.03	DPS AMSE plug-in	0.029	–	–
Pareto light	fraction=0.03	DPS variance	0.029	0.098	–
Pareto light	fraction=0.05	DPS AMSE plug-in	0.050	–	–
Pareto light	fraction=0.05	DPS variance	0.050	0.091	–
Pareto light	fraction=0.1	DPS AMSE plug-in	0.100	–	–
Pareto light	fraction=0.1	DPS variance	0.100	0.083	–
Abs. Student-3	fraction=0.03	DPS AMSE plug-in	0.029	0.116	0.895
Abs. Student-3	fraction=0.03	DPS variance	0.029	0.116	0.894
Abs. Student-3	fraction=0.05	DPS AMSE plug-in	0.050	0.146	0.917
Abs. Student-3	fraction=0.05	DPS variance	0.050	0.146	0.917
Abs. Student-3	fraction=0.1	DPS AMSE plug-in	0.100	0.285	0.905
Abs. Student-3	fraction=0.1	DPS variance	0.100	0.285	0.905

Table 4.7: Target sensitivity for $m = 10$, strong allocation and $k_j/n_j = 0.05$.

DGP	np	Estimator	η_n	Log RMSE	Coverage
Burr heavy	1	DPS AMSE plug-in	0.014	0.282	0.779
Burr heavy	1	DPS variance	0.014	0.282	0.779
Burr heavy	5	DPS AMSE plug-in	0.051	0.221	0.753
Burr heavy	5	DPS variance	0.051	0.221	0.753
Burr heavy	10	DPS AMSE plug-in	0.090	0.207	0.697
Burr heavy	10	DPS variance	0.090	0.207	0.698
Burr light	1	DPS AMSE plug-in	0.359	0.076	0.785
Burr light	1	DPS variance	0.359	0.076	0.785
Burr light	5	DPS AMSE plug-in	0.724	0.069	0.590
Burr light	5	DPS variance	0.724	0.069	0.590
Burr light	10	DPS AMSE plug-in	1.014	0.075	0.428
Burr light	10	DPS variance	1.014	0.075	0.427
Pareto heavy	1	DPS AMSE plug-in	0.014	–	–
Pareto heavy	1	DPS variance	0.014	0.245	–
Pareto heavy	5	DPS AMSE plug-in	0.051	–	–
Pareto heavy	5	DPS variance	0.051	0.204	–
Pareto heavy	10	DPS AMSE plug-in	0.090	–	–
Pareto heavy	10	DPS variance	0.090	0.190	–
Pareto light	1	DPS AMSE plug-in	0.359	–	–
Pareto light	1	DPS variance	0.359	0.087	–
Pareto light	5	DPS AMSE plug-in	0.724	–	–
Pareto light	5	DPS variance	0.724	0.090	–
Pareto light	10	DPS AMSE plug-in	1.014	–	–
Pareto light	10	DPS variance	1.014	0.098	–
Abs. Student-3	1	DPS AMSE plug-in	0.128	0.222	0.924
Abs. Student-3	1	DPS variance	0.128	0.222	0.924
Abs. Student-3	5	DPS AMSE plug-in	0.297	0.142	0.917
Abs. Student-3	5	DPS variance	0.297	0.142	0.917
Abs. Student-3	10	DPS AMSE plug-in	0.441	0.109	0.880
Abs. Student-3	10	DPS variance	0.441	0.109	0.880

4.8 Weight stability and limitations

Table 4.8 reports weight diagnostics in the representative main design. The Daouia–Padoan–Stupfler variance row is the deterministic variance-optimal baseline; there is no separate oracle variance row because, in this common-tail independent design, the oracle and plug-in variance formulas both reduce to $\omega_j = k_j/k$. The oracle and plug-in AMSE weights are stable in this equal-threshold grid: negative weights are absent, and the average maximum absolute weight is close to the largest deterministic allocation weight. The diagnostic matters because it rules out an obvious failure mode. The lack of AMSE improvement in Table 4.1 is not caused by unstable AMSE weights; rather, in the main common-threshold design the AMSE rule is numerically close to the simpler variance allocation.

Table 4.9 gives the AMSE rule a more discriminating environment. The bridge-Hill vector is fixed at $\boldsymbol{\nu}_n^k$, while $\boldsymbol{\omega}$ varies between Daouia–Padoan–Stupfler variance, oracle AMSE, and plug-in AMSE weights. Here AMSE weighting can matter. In Burr light, the oracle AMSE row raises coverage from 0.600 under variance weights to 0.732, while the plug-in AMSE row raises it to 0.642. In the absolute Student model, AMSE weights reduce log RMSE relative to variance weights, and the plug-in AMSE coverage is close to nominal. This gain comes with a warning: the Student AMSE weights are signed in almost every replication. In Burr heavy, AMSE weights give only a small improvement. Exact Pareto plug-in AMSE rows remain omitted for the source reason described above.

The simulation evidence is intentionally limited. It supports the promoted estimator in the iid common-marginal distributed setting and suggests that Daouia–Padoan–Stupfler variance weights are a robust practical baseline for $\boldsymbol{\omega}$. Heterogeneous-threshold diagnostics show that AMSE weights are a genuine alternative worth reporting, but not uniformly superior or uniformly stable. The study also shows where the asymptotic theory should not be overinterpreted: first-order plug-in intervals may under-cover at realistic sample sizes, and the exact decomposition reveals finite-sample contributions from terms that are asymptotically negligible under the theorem's scaling. The study does not address non-identical margins, time series, multivariate targets, random bridge-Hill weights, or lower-order optimality of $\boldsymbol{\nu}$. Those questions would require additional mathematical work before they could become simulation targets.

Table 4.8: Pooled-Weissman weight diagnostics in the representative main design.

DGP	m	Regime	Estimator	Neg. rate	Avg. max $\lvert\omega\rvert$
Burr heavy	10	strong	DPS variance	0.000	0.246
Burr heavy	10	strong	DPS AMSE oracle	0.000	0.246
Burr heavy	10	strong	DPS AMSE plug-in	0.000	0.246
Burr light	10	strong	DPS variance	0.000	0.246
Burr light	10	strong	DPS AMSE oracle	0.000	0.246
Burr light	10	strong	DPS AMSE plug-in	0.000	0.246
Pareto heavy	10	strong	DPS variance	0.000	0.246
Pareto heavy	10	strong	DPS AMSE oracle	0.000	0.246
Pareto heavy	10	strong	DPS AMSE plug-in	–	–
Pareto light	10	strong	DPS variance	0.000	0.246
Pareto light	10	strong	DPS AMSE oracle	0.000	0.246
Pareto light	10	strong	DPS AMSE plug-in	–	–
Abs. Student-3	10	strong	DPS variance	0.000	0.246
Abs. Student-3	10	strong	DPS AMSE oracle	0.000	0.241
Abs. Student-3	10	strong	DPS AMSE plug-in	0.000	0.244

Table 4.9: Pooled-Weissman weight comparison under heterogeneous local threshold fractions. The bridge-Hill weights are fixed at $\boldsymbol{\nu} = \boldsymbol{\nu}_n^k$.

DGP	Estimator	Log bias	Log RMSE	Coverage	Neg. rate	Avg. max $\lvert\omega\rvert$
Burr heavy	DPS AMSE oracle	0.094	0.231	0.755	0.000	0.174
Burr heavy	DPS AMSE plug-in	0.091	0.231	0.767	0.000	0.203
Burr heavy	DPS variance	0.097	0.234	0.744	0.000	0.147
Burr light	DPS AMSE oracle	-0.036	0.067	0.732	0.000	0.174
Burr light	DPS AMSE plug-in	-0.037	0.068	0.642	0.000	0.203
Burr light	DPS variance	-0.034	0.067	0.600	0.000	0.147
Pareto heavy	DPS AMSE oracle	-0.010	0.208	0.746	0.000	0.147
Pareto heavy	DPS AMSE plug-in	–	–	–	–	–
Pareto heavy	DPS variance	-0.010	0.208	–	0.000	0.147
Pareto light	DPS AMSE oracle	-0.072	0.091	0.687	0.000	0.147
Pareto light	DPS AMSE plug-in	–	–	–	–	–
Pareto light	DPS variance	-0.072	0.091	–	0.000	0.147
Abs. Student-3	DPS AMSE oracle	0.068	0.144	0.876	1.000	0.409
Abs. Student-3	DPS AMSE plug-in	0.103	0.142	0.945	0.996	0.264
Abs. Student-3	DPS variance	0.132	0.162	0.923	0.000	0.147

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